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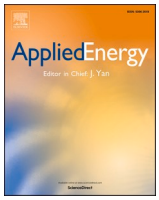

A bi-level planning model of integrated energy capacity configuration for data center cluster considering computing-electricity coupling operation and supply-demand uncertainties

Shuo Zhang [a,c], Shaohui Ding [a], Yingzi Li [b,*], Ming Wei [a], Qinliang Tan [a,c]

[a] School of Economics and Management, North China Electric Power University, Beijing 102206, China
[b] School of Economics and Management, University of Science and Technology Beijing, Beijing 100083, China
[c] Beijing Key Laboratory of New Energy and Low-Carbon Development, North China Electric Power University, Beijing 102206, China

HIGHLIGHTS

- Propose an integrated energy planning framework with computing-electricity coupling.
- Formulate a data-driven DRO bi-level planning model for supply-demand uncertainties.
- Develop a two-stage solution strategy for nested semi-infinite programming.

ARTICLE INFO

Keywords:
Data center cluster
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ABSTRACT

With the rapid expansion of data centers, their energy consumption has also increased substantially. Renewable-integrated energy systems are emerging as a promising paradigm for data center clusters. Through coordinated computing and power dispatch, they effectively address the spatiotemporal mismatch between energy supply and demand, thereby supporting the green and low-carbon development of new-type power system of China. However, supply-demand uncertainties caused by renewable generation and computing workloads increase planning complexity and may result in resource redundancy and operational imbalance. This study develops a two-stage distributionally robust bi-level capacity configuration model for data center cluster that jointly optimizes capacity configuration and computing-electricity coordinated operation. The model characterizes temporal shifting and spatial migration of workloads and establishes a coupling mechanism among computing demand, electricity consumption, cooling demand, and energy supply. To address supply-demand uncertainties, a joint moment-based ambiguity set is constructed to capture the statistical dependence between renewable generation and computing workloads. The upper level minimizes the lifecycle equivalent annual cost for capacity planning, while the lower level minimizes operating cost and net load deviation under the worst-case distributions. Case studies show that incorporating workload flexibility increases renewable energy utilization by 1.18% and reduces total system cost by 5.43% while enhancing planning robustness. Compared with conventional planning methods, the proposed model achieves a more balanced allocation of renewable generation, energy storage, and cooling capacity. The proposed framework improves economic performance, operational reliability, and renewable energy accommodation, providing an effective solution for computing-electricity coordinated planning of low-carbon data center cluster.

1. Introduction

The rapid advancement of the digital economy, cloud computing, and artificial intelligence (AI) has driven an unprecedented expansion of data center (DC) infrastructure worldwide. The widespread deployment of AI foundation models and high performance computing has further accelerated electricity demand, making the sustainable development of DCs a critical challenge for both the information and energy sectors [1]. As one of the fastest growing electricity consumers, DCs have become a major contributor to global energy demand and carbon emissions [2]. To support carbon neutrality targets and mitigate environmental impacts,

* Corresponding author.
E-mail address: liyz@ustb.edu.cn (Y. Li).

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Nomenclature

Abbreviations

DC	Data Center
DCC	Data Center Cluster
CCHP	Combined Cooling, Heating and Power
IT	Information Technology
ETP	Equivalent Thermal Parameter
COP	Coefficient of Performance
PGU	Power Generating Unit
AC	Absorption Chiller
FL	Fixed Load
TSL	Time-Shiftable Load
STL	Space-Transferable Load
QoS	Quality of Service
PUE-R	Renewable Power Usage Effectiveness
MP	Master Problem
SP	Sub-Problem
RO	Robust Optimization
DRO	Distributionally Robust Optimization
C&CG	Column-and-Constraint Generation
KKT	Karush-Kuhn-Tucker
SIP	Semi-Infinite Programming
SDP	Semi-Definite Programming
MISOCP	Mixed-Integer Second-Order Cone Programming

Parameters

P_{grid}^{max}	Maximum transmission capacity of the connection line
R_{grid}^{ramp}	Maximum ramp rate of grid interaction power
$c_{grid,t}$	Time-of-use electricity price during period t
$P_{wind,t}^{avail}$	Theoretical maximum available wind power at period t
$P_{solar,t}^{avail}$	Theoretical maximum available photovoltaic power at period t
$c_{wind,t}^{d}$	Unit agreement price for wind power
$c_{solar,t}^{d}$	Unit agreement price for photovoltaic power
P_{peak}	Rated power of a single server at full load
P_{idle}	Rated power of a single server at idle state
v_{server}	Maximum computing speed of a single server
C_{air}	Specific heat capacity of air
V_i	Effective volume of the server room
R_{wall}	Thermal resistance of the building envelope
δ_{PDU}	Comprehensive power distribution loss coefficient
δ_{loss}	Self-discharge rate of the storage system
η_{ch},η_{dis}	Charging and discharging efficiency
P_{rate}	Maximum charge or discharge power to capacity ratio
$\sigma_{\max},\sigma_{\min}$	Maximum and minimum state-of-charge coefficients
η_{ge},η_{rec}	Electrical and thermal efficiency of PGU
$P_{unit}^{\max}$	Rated power of a single PGU unit
R_{up},R_{down}	Upper and lower ramp rate limits
COP_{AR}	Coefficient of performance of the AC
COP_{HP}	Coefficient of performance of the heat pump
$L_{i,t}^{TSL}$	Arrival volume of TSL during period t
$L_{i,t}^{que,max}$	Maximum allowed backlog queue length for TSL
α_{STL}	Maximum allowable spatial transfer proportion
BW_{max}	Maximum transmission bandwidth limit of the cluster WAN
r	Discount rate
Y	Design lifetime of the DCC
$c_{stor}^{inv},c_{DC}^{inv},c_{CCHP}^{inv}$	Unit configuration costs for storage, servers, and CCHP
$c_{stor}^{O\&M},c_{DC}^{O\&M},c_{CCHP}^{O\&M}$	Unit operation and maintenance costs for storage, servers, and CCHP
$c_{grid,t},c_{wind,t}^{d},c_{solar,t}^{d},c_{gas,t}$	Prices for grid electricity, wind, PV, and natural gas
c_{TSL}	Unit penalty cost for TSL
c_{STL}	Unit penalty cost for STL
c_{Gsert}	Unit price of green certificates
ϕ	Net load deviation penalty coefficient
$\mathscr{D}$	Distributional ambiguity set
S	Number of historical samples
γ_1,γ_2	Robust conservatism parameters
C_{inv}^{max}	Upper limit of financial budget for total investment
$X_{IT,i}^{max},X_{PGU}^{max}$	Maximum allowable installation number of servers and CCHP units
$\alpha_{max},\alpha_{min}$	Upper and lower limits of storage power-to-capacity ratio
η_{RES}^{min}	Threshold of renewable energy consumption weight
β_{pue}	Upper limit standard for PUE-R

Variables

$P_{grid,t}$	Power interaction between the cluster and the external grid during period t
$P_{wind,t}^{used}$	Actual wind power consumed by the system during period t
$P_{solar,t}^{used}$	Actual photovoltaic power consumed by the system during period t
$X_{IT,i}$	Planned total number of configured servers
$P_{IT,i,t}$	Total IT energy consumption during period t
$n_{on,i,t}$	Number of active servers during period t
$\mu_{cpu,i,t}$	Average CPU utilization rate during period t
$Q_{gain,i,t}^{d}$	Heat generated by IT equipment during period t
$Q_{cool,i,t}^{d}$	Cooling load actively removed by the air conditioning system during period t
$Q_{cool,i,t}^{d}$	Heat conduction caused by indoor-outdoor temperature difference during period t
$E_{stor,i}$	Planned capacity for energy storage
$E_{i,t}$	Energy stored in BESS at the end of period t
$P_{ch,i,t},P_{dis,i,t}$	Charging and discharging power at period t
X_{PGU}	Number of configured gas turbines
$P_{gas,t}$	Natural gas power input to PGU at period t
$Q_{rec,t}$	Recovered waste heat power at period t
$Q_{cool,t}^{AR}$	Cooling capacity of the AC at period t
$P_{HP,t}$	Heating or cooling output of the heat pump at period t
$Q_{cool,t}^{HP}$	Auxiliary cooling provided by the water-source heat pump during period t
$L_{arr,i,t}$	Total computing load request arrived at DC i during period t
$L_{i,t}$	Total computing load executed by DC i during period t
$L_{i,t}^{que}$	Backlogged task volume of TSL at DC i at the end of period t
$L_{i,t}^{TSL}$	Executed volume of TSL during period t
$L_{ii',t}$	STL migrated from DC i to i' during period t
$L_{i,t}^{STL}$	Executed volume of STL at DC i during period t
$X_{Gsert,i,t}$	Number of green certificates purchased during period t
x	Stage-I decision variables for capacity planning
y	Stage-II decision variables for operation scheduling
ξ	Random parameters representing multi-source uncertainties
$Q(x,\xi)$	Expected minimum operating cost function
C_{op}	Operational scheduling cost
$\phi\cdot\Delta P_{net}$	Net load deviation penalty term
λ	Lagrangian dual multiplier for probability normalization constraint
ν	Lagrangian dual multiplier for first moment constraint
Γ	Lagrangian dual multiplier for second moment constraint

renewable energy is being increasingly integrated into DC energy systems, promoting the transition of DCs from passive electricity consumers to low-carbon energy hubs [3]. Recent studies indicate that AI oriented and geographically distributed DCs are becoming grid interactive energy consumers with increasingly strong computing-electricity coupling across computing, power, and thermal systems [4]. Meanwhile, the deployment of multi energy technologies, including combined cooling, heating and power (CCHP) systems, energy storage systems, waste heat recovery units, and advanced cooling infrastructures, has turned modern DCs into integrated energy systems with strong interactions among electricity, cooling, and heat [5]. Unlike traditional industrial loads, DC workloads possess inherent temporal and spatial flexibility, allowing computational tasks to be shifted across time periods or migrated among geographically distributed facilities according to renewable energy availability and operating conditions [6]. Such workload flexibility enables DCs to actively participate in demand response and power system operation, creating new opportunities for renewable energy accommodation, grid flexibility enhancement, and carbon emission reduction [7]. Nevertheless, renewable intermittency, increasingly complex multi energy infrastructures, and stronger computing-electricity coupling substantially complicate long-term infrastructure planning. Developing planning frameworks that jointly coordinate renewable integration, workload flexibility, and multi energy resources therefore remains a key challenge for the sustainable development of data center cluster (DCC) [8].

In the area of integrated energy system planning for DCCs, extensive studies have investigated the coordinated planning of renewable generation, CCHP units, energy storage systems, and cooling infrastructures to improve economic performance, energy efficiency, and carbon reduction. Rostirolla et al. [9] reviewed renewable powered DCs and summarized key technologies for improving operational flexibility and renewable energy utilization, while noting the lack of systematic planning methods. Liu et al. [10] developed a low energy consumption DC integrating waste heat recovery, flexible cooling, and hydrogen technologies, but interactions with external integrated energy systems were not explicitly considered. Bian et al. [11] proposed a shared energy storage model for DCCs, while neglecting workload flexibility. He et al. [12] optimized hybrid renewable configurations for DC power supply systems, but modeled workloads as fixed demands. Liu et al. [13] developed a three-stage collaborative scheduling framework for an internet data center and a regional integrated energy system with electricity-heat-water coupling, but focused on operational coordination rather than long-term capacity planning. Du et al. [14] proposed a distributionally robust chance-constrained planning model for regional integrated electricity-heat systems with data centers under wind power uncertainty, but did not characterize the joint dependence between renewable generation and computing workloads. Li et al. [15] proposed a decentralized optimization framework for integrated electricity heat systems with DC based energy hubs, but did not incorporate workload flexibility into planning. Wang et al. [16] developed multi-objective distributionally robust optimization (DRO) planning for hydrogen-involved renewable CCHP systems under source-load uncertainties, but did not model computing workloads or DCC coordination. Zhang et al. [5] and Kumar et al. [2] highlighted the importance of integrating DCs with smart grids and district energy systems, without developing detailed planning models. Overall, existing studies mainly focus on energy infrastructure planning, whereas computing workloads are generally treated as deterministic demands. Consequently, the planning value of workload flexibility and computing-electricity coupling remains insufficiently explored, limiting the coordinated long-term planning of computing and energy resources.

In the area of computing-electricity coupling and workload flexibility, recent studies have recognized geographically distributed DCs as flexible resources that support power system operation through workload migration and demand response. Naoi et al. [17] integrated DC location planning with spatiotemporal workload shifting for fully renewable powered DCs, but focused primarily on operational coordination. Ju et al. [18] developed a two-stage robust transaction and benefit-allocation model for renewable power stations with shared energy storage, green certificates, and virtual energy storage, but did not consider data-center workload shifting. Song et al. [19] developed CloudSimPer to evaluate energy aware workload allocation strategies for geographically distributed renewable powered DCs, although it mainly serves as a simulation platform rather than a planning framework. Zeng et al. [20] proposed a bi-level planning framework incorporating cloud DC flexibility under electricity carbon markets, explicitly modeling workload transferability and curtailment. Recent studies have further emphasized the role of geographically distributed computing resources in power system flexibility. Zhang and Zavala [21] proposed a market mechanism for remunerating spatiotemporal workload shifting, demonstrating its value in providing flexibility services. Cao et al. [7] investigated DCCs as non-wire alternatives in balancing markets and showed that coordinated workload scheduling can relieve network congestion while improving renewable energy utilization. Zhang et al. [22] further explored coordinated demand response between DCs and load aggregators, highlighting the importance of multi agent coordination. Beyond workload migration, Ren et al. [23] proposed a demand response based distributed multi energy system for DCs and demonstrated the benefits of coordinated energy management. Ali A and Özkasap Ö [24] developed thermal aware workload management strategies by jointly optimizing workload allocation and cooling efficiency, reducing cooling energy consumption while maintaining Quality of Service (QoS). Overall, existing studies mainly investigate workload flexibility from an operational perspective, whereas its long-term planning value and interactions with cooling and energy infrastructures remain insufficiently explored. Consequently, workload flexibility and computing-electricity coupling have not been systematically incorporated into long-term integrated energy system planning.

In the area of uncertainty modeling and DRO, existing studies have employed stochastic optimization, robust optimization (RO), and DRO to address renewable generation uncertainty and load fluctuations in integrated energy systems. Esfahani and Kuhn [25] established a Wasserstein metric based DRO framework that has become a representative methodology for data-driven uncertainty modeling. Chen et al. [26] proposed a DRO scheduling model for park level integrated energy systems considering generalized energy storage, but heterogeneous storage degradation was not modeled. Siqin et al. [27] developed a data-driven DRO framework for multi community integrated energy systems with integrated demand response, focusing on short term operation rather than long-term planning. Wang et al. [16] investigated multi objective DRO planning for renewable CCHP systems with hydrogen technologies and demonstrated the effectiveness of moment-based ambiguity sets in managing supply-demand uncertainties. Li et al. [28] proposed a Wasserstein DRO planning model for renewable generation and energy storage under multiple uncertainties, although solution conservativeness depends strongly on the Wasserstein radius and computational complexity increases with scenario size. More recently, Zhang et al. [29] embedded a Wasserstein based two-stage DRO formulation into a bi-level capacity planning framework for a zero carbon park integrated energy system. The proposed framework coordinated long-term capacity planning with short-term robust operation under wind power, photovoltaic generation, and load uncertainties. However, workload flexibility was not considered, and the statistical dependence between renewable generation and computing demand was ignored. Kim et al. [30] introduced a data-driven ambiguity set for integrated energy system planning, but only renewable uncertainty was considered. Zhai et al. [31] developed a data-driven DRO energy management framework with joint chance constraints for multi energy microgrids, focusing on operational decision making. Zhang et al. [32] proposed a data-driven DRO model for peer to peer energy trading, without considering coupled planning decisions. Wang et al. [33] applied DRO to DCC planning by coordinating photovoltaic generation,

energy storage systems, and flexible substations. However, independent uncertainties were assumed, and the dependence between renewable generation and workload flexibility was neglected. Overall, existing studies have advanced DRO methods for integrated energy systems, but most focus on operational scheduling or independently model renewable uncertainty. The spatiotemporal correlations, cross correlations, and statistical dependence between renewable generation and computing workloads remain largely unexplored in long-term planning of computing-electricity coupled systems. Therefore, a unified planning framework incorporating supply-demand uncertainties, workload flexibility, and distributional ambiguity under joint moment information is still lacking.

Despite substantial progress in integrated energy system planning, computing-electricity coupling, and DRO, several research gaps remain. Existing studies primarily focus on energy infrastructure planning while treating computing workloads as exogenous or deterministic demands, leaving the long-term planning value of workload flexibility largely unexplored. Although workload migration has been shown to improve operational performance, its coordination with long-term planning of energy, cooling, and computing resources remains limited. Moreover, few studies explicitly model computing-electricity coupling within a unified planning framework. Existing DRO approaches also rely on simplified uncertainty representations and rarely capture the statistical dependence between renewable generation and computing workloads. Consequently, a unified planning framework integrating computing-electricity coupling, workload flexibility, coordinated energy infrastructure planning, and DRO under supply-demand uncertainties is still lacking.

To address these gaps, this paper proposes a data-driven DRO based integrated energy planning model for DCCs under supply-demand uncertainties. The proposed model jointly optimizes renewable generation, CCHP units, energy storage systems, and computing workload scheduling, while explicitly capturing the statistical dependence between renewable generation and computing workloads. The main contributions are summarized as follows.

(1) An integrated energy capacity configuration planning framework for DCC considering computing-electricity coupling is proposed. Virtual queue theory and multi node network flow models are used to characterize the temporal shifting and spatial migration of workloads, transforming computing tasks into generalized virtual energy flows. A coupled computing electricity thermal model quantifies the planning value of workload flexibility and enables coordinated optimization of energy infrastructure and workload scheduling.
(2) A data-driven DRO bi-level model is formulated to address supply-demand uncertainties. A joint moment information ambiguity set is constructed using first and second order moments and supply-demand covariance information to capture the spatiotemporal dependence between renewable generation and computing workloads. Compared with stochastic programming and RO, the proposed framework achieves a better balance between planning economy and operational robustness.
(3) A two-stage solution strategy is developed for the multi-level nested model with semi-infinite constraints. Strong duality theory reformulates the worst-case distribution problem as a deterministic semidefinite programming problem. The reformulated model is solved using the Column-and-Constraint Generation (C&CG) algorithm, Karush-Kuhn-Tucker (KKT) conditions and the big M method, resulting in an iterative Mixed-Integer Second-Order Cone Programming (MISOCP) formulation for coordinated investment planning and operational scheduling.

2. System architecture and computing-electricity coupling framework

2.1. Physical architecture of the system

The integrated energy system for the DCC constructed in this paper integrates multi energy technologies and computing collaborative mechanisms. It consists of energy supply, energy conversion and

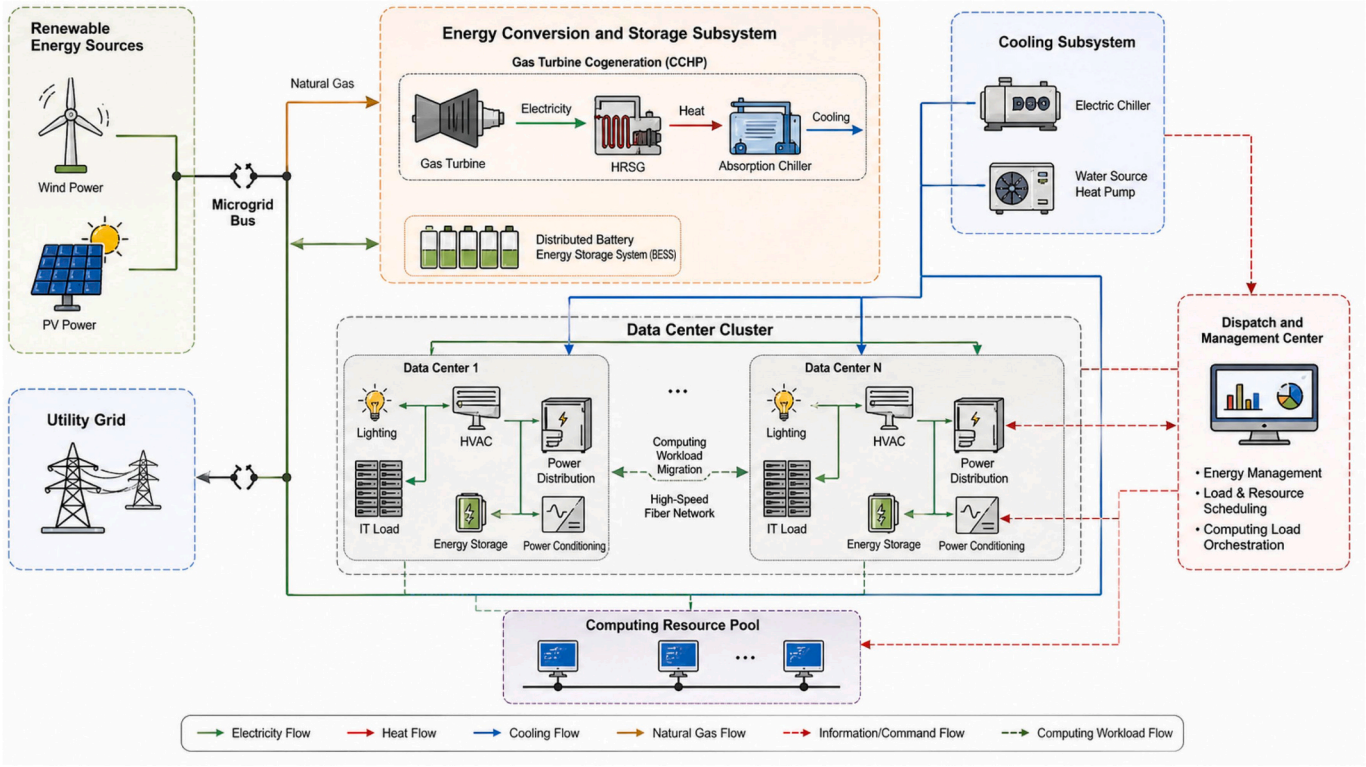

Fig. 1. Physical architecture of the integrated energy system for the DCC.

storage, and computing load subsystems.

On the energy supply side, the system utilizes wind power generation and photovoltaic units as the main primary energy sources. Renewable electricity is directly supplied to DCs through the internal microgrid to enhance local renewable energy consumption. Meanwhile, the system maintains bidirectional power exchange with the utility grid through tie lines. The utility grid provides supplementary power to maintain system balance, forming a supply structure dominated by renewable energy and supported by the main grid.

To improve overall energy efficiency and reduce cooling demand, the system incorporates a regulation subsystem based on CCHP technology. Natural gas is used as the input fuel. Gas turbines generate electricity to supply the base and peak loads of the DCs, while recovered waste heat is converted into cooling energy through absorption chiller (AC). This cascade utilization mode enables coordinated utilization of electricity, heat, cooling, and gas and reduces the electricity consumption of electric chillers. In addition, distributed electrochemical energy storage systems installed in each DC smooth short term renewable fluctuations and enhance operational flexibility through peak shaving and valley filling.

On the load side, the DCC is interconnected through a high speed optical network to form a computing resource pool. The cluster therefore behaves as an energy prosumer with spatiotemporal regulation capability. Based on the real time operating conditions and load levels of each subsystem, the dispatch center coordinates workload migration among DCs and workload shifting over time. Through this physical interconnection, computing flows participate in system regulation as virtual energy carriers, forming an integrated energy system characterized by source grid load storage coordination and deep computing-electricity coupling, as shown in Fig. 1.

2.2. Grid interaction and renewable generation modeling

The power supply system of the DCC consists of the utility grid and local renewable generation units. The utility grid not only compensates for local power deficits but also absorbs surplus renewable electricity. The power exchanged between the DCC and the utility grid is denoted by $P_{grid,t}$ and is allowed to flow bidirectionally. Owing to transformer capacity and tie line transmission limits, the exchanged power must remain within the allowable range and satisfy ramping constraints to avoid adverse impacts on the utility grid:

$$\begin{cases} |P_{grid,t}| \le P_{grid}^{max} \\ |P_{grid,t} - P_{grid,t-1}| \le R_{grid}^{ramp} \end{cases} \tag{1}$$

where P_{grid}^{max} is the maximum transmission capacity; R_{grid}^{ramp} is the maximum ramp rate of grid interaction power.

From an economic perspective, the grid interaction cost is determined by the time of use electricity tariff. The electricity price vector $c_{grid,t}$ incentivizes electricity purchase and energy storage charging during off peak periods and promotes storage discharging and reduced grid purchases during peak periods.

The local renewable generation system includes wind turbines and photovoltaic arrays. Their outputs exhibit significant uncertainty and variability due to meteorological conditions. In this paper, the maximum available wind and photovoltaic power are regarded as uncertain parameters and are denoted by $P_{wind,t}^{avail}$ and $P_{solar,t}^{avail}$, respectively. Because of power balance and renewable accommodation limits, the actual renewable power injected into the system may be lower than the available power. Therefore, the actual accommodated wind and photovoltaic power denoted by $P_{wind,t}^{used}$ and $P_{solar,t}^{used}$, which are introduced as decision variables and satisfy:

$$\begin{cases} 0 \le P_{wind,t}^{used,d} \le P_{wind,t}^{avail,d} \\ 0 \le P_{solar,t}^{used,d} \le P_{solar,t}^{avail,d} \end{cases} \tag{2}$$

This constraint allows renewable curtailment under extreme operating conditions to maintain system security. To quantify renewable utilization costs and encourage renewable consumption, unit electricity purchase prices for wind and photovoltaic power, $c_{wind,t}^{d}$ and $c_{solar,t}^{d}$, are introduced. These costs, together with grid interaction costs, constitute the operating energy expenditure of the system.

2.3. Energy consumption and thermodynamic modeling of DCs

The energy consumption of DC mainly consists of information technology (IT) equipment, cooling systems, power distribution and lighting systems. The energy consumption of cooling and auxiliary systems strongly depends on IT power consumption [34]. Based on the coupling between energy flows and information flows, the following mathematical models are established.

2.3.1. IT equipment energy consumption model

IT equipment constitutes the core load of a DC. Its power consumption is positively correlated with the number of active servers and CPU utilization. Assuming homogeneous server specifications, which is consistent with the standardized deployment architecture of modern cloud DCs [35], the IT power consumption during period t is expressed as:

$$P_{IT,i,t} = n_{on,i,t}\left[P_{idle} + \left(P_{peak} - P_{idle}\right)\mu_{cpu,i,t}\right] + n_{off,i,t}P_{sleep} \tag{3}$$

where $n_{on,i,t}$ is the number of active servers; P_{peak} and P_{idle} are the server power under full load and idle conditions, respectively; $\mu_{cpu,i,t}$ is the average CPU utilization. Following the widely adopted linear server power model in [36], server power consumption is assumed to vary linearly with CPU utilization. Therefore, to reduce the computational complexity, this study reasonably assumes $P_{sleep} = 0$.

The CPU utilization is related to the processed computing workload $L_{i,t}$ through

$$\mu_{cpu,i,t} = \frac{L_{i,t}}{n_{on,i,t}v_{server}} \tag{4}$$

Substituting this relationship into (3) eliminates the nonlinear terms and yields the simplified IT power model:

$$\begin{aligned} P_{IT,i,t} &= n_{on,i,t}P_{idle} + \left(P_{peak} - P_{idle}\right)\frac{L_{i,t}}{v_{server}} \\ s.t. &\frac{L_{i,t}}{v_{server}} \le n_{on,i,t} \le X_{IDC,i} \end{aligned} \tag{5}$$

where v_{server} is the maximum processing speed of a single server; $X_{IT,i}$ is the total number of servers installed in DC i.

2.3.2. Air conditioning system energy consumption model

The thermal environment of a DC exhibits significant thermal inertia. To exploit the flexibility of temperature regulation, a first order equivalent thermal parameter (ETP) model is adopted to describe the indoor temperature dynamics. Since ambient temperature varies considerably across seasons, thermal variables are defined for representative seasonal days. According to the law of energy conservation, the discretized thermal balance equation of the server room is:

$$\begin{aligned} T_{in,i,t+1} &= T_{in,i,t} + \frac{\Delta t}{C_{air}\cdot V_i}\left(Q_{gain,i,t}^{d} - Q_{cool,i,t}^{d} - Q_{loss,i,t}^{d}\right) \\ s.t. &\; T_{\min} \le T_{in,i,t} \le T_{\max} \end{aligned} \tag{6}$$

where C_{air} is the specific heat capacity of air; V_i is the effective room volume; $Q^d_{gain,i,t}$ is the heat generated by IT equipment and is approximated by $\beta_{heat}P_{IT,i,t}$ ($\beta_{heat} \approx 1$); $Q^d_{cool,i,t}$ is the cooling capacity provided by the air conditioning system; $Q^d_{loss,i,t}$ is the heat transfer caused by the indoor and outdoor temperature difference is expressed as $\left(T_{in,i,t} - T^d_{out,t}\right)/R_{wall}$, where R_{wall} denotes the thermal resistance of the building envelope. To ensure safe operation of IT equipment, the server room temperature $T_{in,i,t}$ must remain within a prescribed range. The thermal inertia of indoor air can therefore be exploited to temporarily store heat by reducing cooling power during periods of high electricity prices or insufficient renewable generation, reflecting the flexibility arising from computing-electricity coupling.

The electrical power consumption $P^d_{AC,i,t}$ and cooling capacity $Q^d_{cool,i,t}$ of the air conditioning system are related through the coefficient of performance (COP):

$$P^d_{AC,i,t} = \frac{Q^d_{cool,i,t}}{COP_{i,t}} \tag{7}$$

2.3.3. Auxiliary system energy consumption modeling

Power distribution losses and lighting consumption are relatively stable and are modeled as linear auxiliary losses associated with the main load $P_{IT,i,t}$. Under a representative day of season d, the total power consumption $P^d_{DC,i,t}$ of DC i during period t is expressed as:

$$P^d_{DC,i,t} = (1 + \delta_{PDU}) \cdot \left(P_{IT,i,t} + P^d_{AC,i,t}\right) \tag{8}$$

where δ_{PDU} is the comprehensive power loss coefficient, including lighting consumption, transformer losses, and UPS losses.

2.4. Modeling of key equipment for multi energy complementation

The integrated energy system of the DCC considered in this paper comprises distributed electrochemical energy storage systems and a centralized CCHP system.

2.4.1. Modeling of the energy storage system

To smooth renewable generation fluctuations and exploit electricity price arbitrage [37], an electrochemical energy storage system is installed in each DC. Its operating state is characterized by the state of charge, charging and discharging power, and logical constraints. The energy evolution satisfies the conservation law:

$$E_{i,t} = (1 - \delta_{loss})E_{i,t-1} + \left(\eta_{ch}P_{ch,i,t} - \frac{P_{dis,i,t}}{\eta_{dis}}\right)\Delta t \tag{9}$$

where $E_{i,t}$ is the electricity stored of the energy storage system at the end of period t; δ_{loss} is the self-discharge rate; $P_{ch,i,t}$ and $P_{dis,i,t}$ are the charging and discharging powers, respectively; η_{ch} and η_{dis} are the charging and discharging efficiencies, respectively.

To prevent simultaneous charging and discharging and to enforce power and energy limits, binary variables $u_{ch,i,t}$ and $u_{dis,i,t}$ are introduced:

$$\begin{cases} 0 \le P_{ch,i,t} \le u_{ch,i,t}P_{rate}E_{stor,i} \\ 0 \le P_{dis,i,t} \le u_{dis,i,t}P_{rate}E_{stor,i} \\ u_{ch,i,t} + u_{dis,i,t} \le 1 \\ \sigma_{\min}E_{stor,i} \le E_{i,t} \le \sigma_{\max}E_{stor,i} \end{cases} \tag{10}$$

where P_{rate} is the charging rate coefficient, representing the ratio between power and energy capacity; $\sigma_{\min}$ and $\sigma_{\max}$ are the minimum and maximum state of charge coefficients used to prolong battery lifetime; $E_{stor,i}$ is the installed energy storage capacity of DC i.

2.4.2. Modeling of the CCHP system

The CCHP system consists of a power generating unit (PGU), a waste heat recovery unit, and an absorption chiller (AC). It enables cascaded utilization of natural gas, electricity, waste heat, and cooling energy to satisfy the base electrical and cooling demands of the DCC.

(1) Gas Power Generation and Heat Production Model

It is assumed that all gas turbines in the CCHP system are identical. During period t, the PGU consumes natural gas to produce electricity and simultaneously releases high temperature waste heat. Under the constant efficiency assumption, the electricity and heat outputs are expressed as [38]:

$$P_{PGU,t} = \eta_{ge}P_{gas,t} \tag{11}$$

$$Q_{rec,t} = \eta_{rec}P_{gas,t} = \frac{\eta_{rec}}{\eta_{ge}}P_{PGU,t} \tag{12}$$

where $P_{gas,t}$ is the power value of the input natural gas converted according to its heating value; $Q_{rec,t}$ is the recovered effective waste heat power; η_{ge} is the power generation efficiency; η_{rec} is the waste heat recovery efficiency.

The output of the PGU is constrained by its rated capacity and ramping limits:

$$\begin{cases} 0 \le P_{PGU,t} \le X_{PGU}P^{\max}_{unit} \\ -R_{down} \le P_{PGU,t} - P_{PGU,t-1} \le R_{up} \end{cases} \tag{13}$$

where X_{PGU} is the number of gas turbines configured in the system; $P^{\max}_{unit}$ is the rated power of a single unit; R_{up} and R_{down} are the upper and lower limit constraints of the ramp rate.

(2) Absorption Refrigeration Model

The AC utilizes waste heat from the PGU to provide cooling, thereby reducing electricity consumption by electric chillers. During period t, the relationship between cooling output $Q^{AR}_{cool,t}$ and heat input $Q^{AR}_{in,t}$ is given by [39]:

$$\begin{aligned} &Q^{AR}_{cool,t} = COP_{AR}Q^{AR}_{in,t} \\ &s.t. 0 \le Q^{AR}_{in,t} \le Q_{rec,t} \end{aligned} \tag{14}$$

where COP_{AR} is the COP of the AC. The heat input cannot exceed the total recovered waste heat, and the remaining heat is treated as dissipation loss.

(3) Auxiliary Heat Pump Model

A water source heat pump is further installed to recover low grade heat and provide auxiliary cooling. Its operating model is given by:

$$Q^{HP}_{cool,t} = COP_{HP}P_{HP,t} \tag{15}$$

where $P_{HP,t}$ is the electrical consumption of the heat pump; $Q^{HP}_{cool,t}$ is its cooling or heating output.

2.5. Spatiotemporal scheduling mechanism of computing workloads

The spatiotemporal scheduling of computing workloads is essential for realizing computing-electricity coupling. According to the latency requirements of different services, the computing workload requests $L_{arr,i,t}$ arriving at DC i during period t are divided into three categories: Fixed Load (FL), Time-Shiftable Load (TSL), and Space-Transferable Load (STL).

$$L_{arr,i,t} = L_{arr,i,t}^{FL} + L_{arr,i,t}^{TSL} + L_{arr,i,t}^{STL} \tag{16}$$

Computing workloads exhibit limited seasonal variation, and the workload pattern within each typical day is assumed to remain unchanged. Therefore, annual workload arrivals are not differentiated across typical days. Through workload scheduling, the actual executed workload $L_{i,t}$ reconstructs the electricity demand profile of each DC.

2.5.1. Modeling of time-shiftable load

Tasks such as big data processing and model training can be shifted over time and exhibit high latency tolerance. A virtual queue model is adopted to characterize their temporal flexibility. Each DC is regarded as a buffer with computing storage capability, where tasks can be temporarily accumulated and processed when renewable generation is abundant or electricity prices are low. The queue dynamics satisfy:

$$L_{i,t}^{que} = L_{i,t-1}^{que} + L_{arr,i,t}^{TSL} - L_{i,t}^{TSL} \tag{17}$$

where $L_{i,t}^{que}$is the task backlog of DC i at the end of period t; $L_{i,t}^{TSL}$ is the arrival volume of TSL tasks during period t; $L_{i,t}^{TSL}$ is the processed volume of TSL tasks during period t. To guarantee QoS and avoid unlimited backlog accumulation, the following constraints are imposed:

$$\begin{cases} 0 \le L_{i,t}^{que} \le L_{i,t}^{que,max} \\ L_{i,0}^{que} = L_{i,T}^{que} = 0 \\ 0 \le L_{i,t}^{TSL} \le L_{arr,i,t}^{TSL} + L_{i,t-1}^{que} \end{cases} \tag{18}$$

where $L_{i,t}^{que,max}$ is the maximum allowable queue length, reflecting the latency tolerance of this type of service. The initial and final queue lengths are both set to zero to ensure that all arriving tasks are processed within the scheduling horizon.

2.5.2. Modeling of space-transferable load

Tasks such as distributed computing and content distribution can be migrated among geographically distributed DCs through high speed optical communication networks. A multi node network flow model is employed to represent spatial workload migration, where computing flows are treated as virtual energy flows. The actual processed STL of DC i is equal to the local arrivals plus the net migrated workload [40]:

$$L_{i,t}^{STL} = L_{arr,i,t}^{STL} + \sum_{i' \in \Omega_{DC}, i' \ne i} \left(L_{i'i,t}^{STL} - L_{ii',t}^{STL} \right) \tag{19}$$

where $L_{i,t}^{STL}$ is the processed volume of STL tasks during period t; $L_{ii',t}^{STL}$ is the workload transferred from DC i to DC i' during period t.

Spatial scheduling is constrained by communication bandwidth and migration capability:

$$\begin{cases} 0 \le \sum_{i' \in \Omega_{DC}, i' \ne i} L_{ii',t}^{STL} \le \alpha_{STL} L_{arr,i,t}^{STL} \\ \sum_{i \in \Omega_{DC}} \sum_{i' \in \Omega_{DC}, i' \ne i} L_{ii',t}^{STL} \le BW_{\max} \end{cases} \tag{20}$$

where $\alpha_{STL} \in [0,1]$ is the maximum transferable proportion, reflecting data localization constraints, and BW_{max} is the transmission capacity of the wide area network.

2.5.3. Computing-electricity coupling response mechanism

For FL such as online transactions and real-time interactive services, are highly sensitive to latency. These tasks must be processed immediately and cannot be delayed or migrated. Therefore, the processed FL of DC i is equal to its arrival volume during period $L_{i,t}^{FL} = L_{arr,i,t}^{FL}$. Combining the scheduling mechanisms of the three workload categories, the total workload $L_{i,t}$ executed by DC i during period t is expressed as:

$$L_{i,t} = L_{i,t}^{FL} + L_{i,t}^{TSL} + L_{i,t}^{STL} \tag{21}$$

This variable is directly substituted into the IT electricity consumption model (5) in Section 2.3.1. By optimizing $L_{i,t}^{TSL}$ and $L_{ii',t}^{STL}$, the model coordinates computing workload scheduling with renewable generation fluctuations and realizes computing-electricity coupling. To avoid service degradation and operational oscillations caused by excessive scheduling, a workload scheduling penalty is introduced in the objective function [41]. The resource consumption associated with temporal and spatial workload scheduling is converted into an economic cost and incorporated into the optimization model.

2.6. Analysis of supply-demand dual uncertainty characteristics

The safe and economical operation of the integrated energy system for the DCC relies on effective management of supply-demand uncertainties. On the supply side, wind and photovoltaic generation depend on meteorological conditions and exhibit strong intermittency and temporal variability. In particular, the anti-peak shaving characteristic of wind power often leads to a temporal mismatch with daytime computing demand. On the demand side, although conventional online services exhibit periodic patterns, batch computing tasks such as high performance computing are highly stochastic and bursty, causing sudden fluctuations in computing workloads. The interaction between renewable uncertainty and workload fluctuations can disrupt the balance of multi energy flows and increase the risks of renewable curtailment and local energy shortages.

Traditional deterministic planning methods cannot effectively cope with multi source uncertainties. Stochastic programming can characterize parameter fluctuations, but it relies on accurate probability distributions that are often unavailable in practice. Estimation errors in probability distributions may lead to distorted planning decisions. Robust optimization guarantees system security by considering worst-case scenarios. However, it ignores statistical information and usually produces overly conservative capacity configurations.

To overcome these limitations, this paper adopts DRO. Instead of assuming an exact probability distribution, historical data is used to extract first and second order statistical information and construct a data-driven ambiguity set. Supply-demand covariance information is further incorporated to capture the dependence structure between renewable generation and computing workloads. Although no direct physical coupling exists between them, common environmental conditions and operational patterns may induce statistical dependence. Ignoring this dependence may distort the tail behavior of the joint distribution and underestimate system risks under extreme conditions. Therefore, the proposed ambiguity set seeks optimal investment and scheduling decisions under the worst-case probability distribution while maintaining a reasonable tradeoff between planning economy and operational robustness.

3. Two-stage distributionally robust bi-level planning model

With the scale expansion of DCC and the increasing penetration of renewable energy, DCs are evolving from conventional electricity consumers into flexible prosumers. To address supply-demand uncertainties and computing-electricity coupling, this paper develops a two-stage planning framework that coordinates long-term capacity planning and short term operation. The problem is formulated as a two-stage distributionally robust bi-level optimization model with a Master-Sub problem structure to balance long-term investment economy and operational robustness.

3.1. Problem description and time scales of coordinated optimization

The proposed system is a computing-electricity coupled energy system for a DCC integrating wind power, photovoltaic generation, and multi energy technologies. It exhibits two key characteristics. First, the

system is subject to supply-demand uncertainties arising from wind power, photovoltaic generation, and computing workloads. Second, DC workloads exhibit workload flexibility through temporal shifting and spatial migration, enabling participation in peak shaving, load balancing, and renewable energy accommodation.

3.1.1. Coordinated optimization objective system

The model simultaneously considers economic performance and operational stability. At the economic level, the objective is to minimize the equivalent annual total cost C_{total} over the system lifetime. The total cost includes annualized investment cost C_{inv} of energy storage systems, servers, and CCHP units, operation and maintenance cost $C_{O\&M}$, electricity purchasing cost C_{energy}, green certificate purchasing cost C_{policy}, and workload scheduling penalty cost $C_{schedule}$. The workload scheduling penalty quantifies service quality degradation caused by TSL and communication overhead caused by STL.

$$\min C_{total} = C_{plan} + \mathbb{E}(C_{op}) \tag{22}$$

$$C_{plan} = C_{inv} + C_{O\&M} \tag{23}$$

$$C_{inv} = \frac{r(1+r)^Y}{(1+r)^Y - 1}\left[\sum_{i=1}^{N}\left(E_{stor,i}c_{stor}^{inv} + X_{IT,i}c_{IDC}^{inv}\right) + X_{PGU}c_{CCHP}^{inv}\right] \tag{24}$$

$$C_{O\&M} = \sum_{i=1}^{N}\left(E_{stor,i}c_{stor}^{O\&M} + X_{IT,i}c_{IDC}^{O\&M}\right) + X_{PGU}c_{CCHP}^{O\&M} \tag{25}$$

$$C_{op} = C_{energy} + C_{schedule} + C_{policy} \tag{26}$$

$$C_{energy} = \sum_{d=1}^{D}\left\{\left[\sum_{t=1}^{T}\left(P_{grid,t}c_{grid,t} + P_{wind,t}^{d}c_{wind,t}^{d} + P_{solar,t}^{d}c_{solar,t}^{d} + P_{gas,t}c_{gas,t}\right)\right] \times n_d\right\} \tag{27}$$

$$C_{schedule} = \sum_{i=1}^{N}\sum_{t=1}^{T}\left(c_{TSL}L_{i,t}^{que} + c_{STL}\sum_{i'\neq i}L_{ii',t}^{STL}\right) \tag{28}$$

$$C_{policy} = \sum_{i=1}^{N}\sum_{d=1}^{D}\sum_{t=1}^{T}X_{Gsert,i,t}c_{Gsert}n_d \tag{29}$$

where r is the discount rate; Y is the project lifetime of the DCC; c_{stor}^{inv}, c_{DC}^{inv}, and c_{CCHP}^{inv} are the unit investment costs of energy storage systems, servers, and CCHP units, respectively; $c_{stor}^{O\&M}$, $c_{DC}^{O\&M}$, and $c_{CCHP}^{O\&M}$ are the corresponding unit operation and maintenance costs; $c_{grid,t}$, $c_{wind,t}^{d}$, $c_{solar,t}^{d}$, and $c_{gas,t}$ are the electricity prices of the utility grid, wind power, photovoltaic generation, and natural gas during period t, respectively; c_{TSL} is the unit penalty of TSL, representing the service quality degradation caused by delayed task execution; $c_{STL}$is the unit penalty of STL, representing the communication overhead and inter DC transmission cost caused by cross regional workload migration; $X_{Gsert,i,t}$ is the number of green certificates purchased by DC i during period t; c_{Gsert} is the corresponding unit price.

At the stability level, the objective is to minimize the average absolute deviation of the net load ΔP_{net}. Reducing net load fluctuations improves peak shaving capability and enhances grid friendliness under high renewable penetration. To eliminate dimensional inconsistency in the multi objective formulation, the net load deviation is converted into an economic penalty using the penalty coefficient, resulting in a unified single objective optimization model.

$$\min\Delta P_{net} = \frac{1}{T}\sum_{d=1}^{D}\sum_{t=1}^{T}\left|P_{grid,t}^{d} - P_{grid,ave}^{d}\right| \tag{30}$$

$$P_{grid,t}^{d} = \left(\sum_{i=1}^{N}P_{DC,i,t}^{d} + P_{HP,t}^{d}\right) - \left[P_{wind,t}^{d} + P_{solar,t}^{d} + P_{PGU,t} + \sum_{i=1}^{N}\left(P_{dis,i,t} - P_{ch,i,t}\right)\right] \tag{31}$$

$$P_{grid,ave}^{d} = \frac{1}{T}\sum_{t=1}^{T}P_{grid,t}^{d}, d = 1,\ldots,D \tag{32}$$

where the net load is defined as the difference between the total electricity demand of all DCs and water source heat pumps and the power supplied by wind power, photovoltaic generation, the utility grid, CCHP units, and energy storage systems.

3.1.2. Time scale division and two-stage decision framework

Although the above objectives represent different physical aspects, optimization follows a two-stage decision process with planning preceding operation.

Stage-I corresponds to long-term capacity planning over the project lifetime. The first stage decision variables are $x = \left[E_{stor,i}, X_{DC,i}, X_{PGU}\right]$, including the capacities of energy storage systems, servers, and CCHP units. These decisions are determined before uncertainty realization and provide the physical foundation for subsequent operation.

Stage-II corresponds to day ahead operational scheduling. The second stage decision variables are $y = \left[P_{grid,t}, P_{dis,i,t}, P_{ch,i,t}, P_{PGU,t}, X_{Gsert,i,t}, L_{i,t}^{TSL}, L_{ii',t}^{STL}\right]$, including grid power exchange, energy storage charging and discharging, gas turbine output, green certificate purchasing, and TSL and STL scheduling. After the uncertain parameter including wind power, photovoltaic generation, and computing workloads is realized, the operating strategy is optimized within the capacity limits determined in Stage I. Accordingly, the problem is formulated as the following Min-Max-Min two-stage DRO model.

$$\min_{x\in\mathscr{X}}\left\{C_{inv}(x) + \max_{P\in\mathscr{D}}\mathbb{E}_P[Q(x,\xi)]\right\}$$
$$s.t.\begin{cases}\chi = \{x|(43)\}\\ \Omega(x,\xi) = \{y|(1),(10),(13),(18),(20),(44)\}\end{cases} \tag{33}$$

where $\mathscr{X}$ and $\Omega(x,\xi)$ are the feasible regions of the first and second stage decision variables, respectively; $\mathscr{D}$ is the ambiguity set describing supply-demand uncertainties, which is introduced in Section 3.2; $Q(x,\xi)$ is the second stage recourse function, representing the minimum operating cost under planning decision x and the worst-case probability distribution ξ, defined as follows:

$$Q(x,\xi) = \min_{y\in\Omega(x,\xi)}\left[C_{op}(y) + \phi\cdot\Delta P_{net}(y)\right] \tag{34}$$

where C_{op} is the operating cost; $\phi\cdot\Delta P_{net}$ is the net load deviation penalty term, and ϕ is the penalty coefficient. By incorporating the penalty term into the objective function, the stability objective is transformed into an economic objective, yielding a unified single objective optimization model.

3.2. Uncertainty modeling and DRO framework

3.2.1. Description of supply-demand dual side uncertainty

The planning of a DCC is affected by supply-demand uncertainties from wind power, photovoltaic generation, and computing workloads. These variables exhibit significant stochastic fluctuations and varying statistical characteristics under different weather conditions and workload patterns. Wind power and photovoltaic generation are naturally correlated because they are driven by common meteorological conditions. Although computing workloads are mainly determined by user demand, they may also exhibit statistical dependence on renewable generation under specific climatic and regional operating conditions.

Therefore, assuming independent uncertain variables may distort the joint uncertainty and lead to suboptimal planning decisions. To address this issue, a DRO framework is adopted. An ambiguity set containing all probability distributions satisfying the prescribed statistical information is constructed to explicitly capture distributional uncertainty.

3.2.2. Construction of ambiguity set based on moment information

To characterize the joint uncertainty of renewable generation and computing workloads, a full period random vector consisting of wind power output, photovoltaic generation, and computing workload is defined as:

$$\xi = \left[P_{wind}^{\top}, P_{solar}^{\top}, L^{\top}\right]^{\top} \in \mathbb{R}^{N_{\xi}} \tag{35}$$

$$P_{wind} = \left[P_{wind,1}^{1}, \ldots, P_{wind,T}^{D}\right]^{\top} \tag{36}$$

$$P_{solar} = \left[P_{solar,1}^{1}, \ldots, P_{solar,T}^{D}\right]^{\top} \tag{37}$$

$$L = \left[L_{1,1}, \ldots, L_{N,T}\right]^{\top} \tag{38}$$

where each component is a time series vector with length T, and $N_{\xi} = T \times (2 \times D + N)$ denotes the total dimension of the random vector. The constructed random vector preserves both temporal correlations within each variable and cross correlations among renewable generation and computing workloads.

Since the true probability distribution $\mathbb{P}$ is unavailable and difficult to estimate accurately, a data-driven ambiguity set is constructed using historical statistical information. The ambiguity set $\mathscr{D}$ contains all probability distributions whose first and second moments satisfy the prescribed constraints. The covariance matrix preserves both temporal correlations and cross correlations without imposing explicit assumptions on the probability distribution. The ambiguity set is defined as:

$$\mathscr{D} = \left\{ \mathbb{P} \in \mathscr{P}_0(\Xi) \,|\, \begin{array}{c} \mathbb{P}(\xi \in \Xi) = 1 \\ \mathbb{E}_{\mathbb{P}}[\xi] = \mu_0 \\ \mathbb{E}_{\mathbb{P}}\left[(\xi - \mu_0)(\xi)^{\top}\right] \leq \gamma_1 \Sigma_0 \end{array} \right\} \tag{39}$$

$$\Xi = \left\{ \xi \in \mathbb{R}^{N_{\xi}} \left| \begin{array}{l} \xi_{\min} \leq \xi \leq \xi_{\max} \\ |\xi - \mu_0| \leq \gamma_2 \sigma \end{array} \right. \right\} \tag{40}$$

where μ_0 and Σ_0 are the empirical mean vector and covariance matrix calculated based on historical data; σ is the standard deviation vector; $\gamma_1 \geq 1$ and $\gamma_2 > 0$ are conservativeness parameters that determine the size of the ambiguity set according to the confidence level of historical data. γ_1 controls the allowable deviation of the second moment from its empirical estimate. Larger values of γ_1 enlarge the ambiguity set and lead to more conservative planning decisions. γ_2 limits the distance between feasible realizations and the empirical mean and excludes physically unrealistic extreme scenarios [32].

The empirical mean vector and covariance matrix are estimated from historical samples as follows:

$$\mu_0 = \frac{1}{S} \sum_{s=1}^{S} \widehat{\xi}_s \tag{41}$$

$$\Sigma_0 = \frac{1}{S} \sum_{s=1}^{S} (\widehat{\xi}_s - \mu_0)(\widehat{\xi}_s - \mu_0)^{\top} \tag{42}$$

$$\Sigma_0 = \begin{bmatrix} \Sigma_{WW} & \Sigma_{WP} & \Sigma_{WL} \\ \Sigma_{PW} & \Sigma_{PP} & \Sigma_{PL} \\ \Sigma_{LW} & \Sigma_{LP} & \Sigma_{LL} \end{bmatrix} \tag{43}$$

where $S$is the number of historical samples; Σ_{WW}, Σ_{PP} and Σ_{LL} are the autocovariance matrices of wind power, photovoltaic generation, and computing workloads; Σ_{WP} is the cross covariance matrix between wind power and photovoltaic generation; Σ_{WL} and Σ_{PL} are the potential statistical dependence between renewable generation and computing workloads.

The proposed ambiguity set preserves the first and second order moment information while retaining the dependence structure embedded in historical data. It captures distributional uncertainty caused by forecasting errors and estimation uncertainty without sacrificing computational tractability.

3.3. Model constraints

Based on the physical architecture and uncertainty modeling described above, this section defines the constraints governing the planning and operation of the DCC integrated energy system. These constraints define the feasible regions of the first stage and second stage models, covering investment planning, multi energy balance, and green service requirements.

3.3.1. Planning level constraints

First stage investment decisions must satisfy both physical resource limitations and the available investment budget. The total investment cost cannot exceed the prescribed budget limit C_{inv}^{max}. In addition, equipment installation is constrained by the available site area and load bearing capacity. To ensure proper coordination between power and energy resources, the installed energy storage capacity $E_{stor,i}$ must remain proportional to the IT capacity $X_{IT,i}$. The corresponding constraints are as follows:

$$\begin{cases} \sum_{i=1}^{N} \left(E_{stor,i} c_{stor}^{inv} + X_{IT,i} c_{IDC}^{inv}\right) + X_{PGU} c_{CCHP}^{inv} \leq C_{inv}^{max} \\ X_{IT,i}^{min} \leq X_{IT,i} \leq X_{IT,i}^{max} \\ \alpha_{\min} P_{peak} X_{IT,i} \leq E_{stor,i} \leq \alpha_{\max} P_{peak} X_{IT,i} \\ 0 \leq X_{PGU} \leq X_{PGU}^{max} \end{cases} \tag{44}$$

where $X_{IT,i}^{max}$ and X_{PGU}^{max} are the maximum allowable numbers of installed IT equipment and CCHP gas generator units, respectively; α_{min} and α_{max} are the lower and upper bounds of the power to energy ratio, ensuring sufficient discharge duration for peak shaving and valley filling.

3.3.2. System multi energy balance constraints

During intraday operation, the system must satisfy the supply-demand balance of electricity, heating, and cooling. The electrical power balance requires that total supply equals total demand. The net grid power $P_{grid,t}^{d}$ is adopted, where positive and negative values represent electricity purchase and sale, respectively.

$$P_{grid,t}^{d} + P_{wind,t}^{d} + P_{solar,t}^{d} + \sum_{i=1}^{N} \left(P_{dis,i,t}^{d} - P_{ch,i,t}^{d}\right) + P_{PGU,t}^{d} = \sum_{i=1}^{N} P_{DC,i,t}^{d} + P_{HP,t}^{d} \tag{45}$$

The cooling balance requires that the recovered waste heat from the CCHP unit $Q_{rec,t}$, the cooling output of the electric chiller, and the cooling output of the AC jointly satisfy the cooling demand $Q_{cool,i,t}^{req}$. Pipeline heat losses are neglected, and the balance relationship is as follows:

$$\sum_{i=1}^{N} Q_{cool,i,t}^{d} + Q_{cool,t}^{AR} + Q_{cool,t}^{HP} = \sum_{i=1}^{N} Q_{cool,i,t}^{req} \tag{46}$$

where $Q_{cool,i,t}^{d}$ and $Q_{cool,t}^{AR}$ are the cooling outputs of the electric chiller and AC, respectively; $Q_{cool,t}^{HP}$ is the auxiliary cooling provided by the water source heat pump. This constraint ensures that heat generated by computing workloads is removed promptly, maintaining stable operating conditions in DC.

3.3.3. Green low-carbon metrics and quality of service constraints

To support low-carbon operation, the model incorporates two performance metrics, namely the renewable energy accommodation ratio and PUE-R. The renewable energy accommodation ratio η_{RES}^{min} is constrained such that the combined contribution of locally consumed renewable electricity and purchased green certificates is no less than the prescribed threshold over the planning horizon.

$$\frac{\sum_{d=1}^{D}\sum_{t=1}^{T}\left(P_{wind,t}^{d}+P_{solar,t}^{d}+X_{Gsert,i,t}^{d}\right)}{\sum_{i=1}^{N}\sum_{d=1}^{D}\sum_{t=1}^{T}P_{DC,i,t}^{d}}\geq\eta_{RES}^{\min} \tag{47}$$

The PUE-R constraint limits the ratio of nonrenewable electricity consumption to IT energy consumption below the prescribed threshold β_{pue}, encouraging the priority use of clean energy.

$$\frac{\sum_{t=1}^{T}\left(P_{grid,t}^{buy}+P_{PGU,t}\right)}{\sum_{i=1}^{N}\sum_{t=1}^{T}P_{IT,i,t}}\leq\beta_{pue} \tag{48}$$

3.4. Model reformulation based on duality theory

The proposed two-stage distributionally robust planning model exhibits a nested Min-Max-Min structure. The inner maximization searches for the worst-case probability distribution within the ambiguity set $\mathscr{D}$, resulting in an infinite dimensional optimization problem. To improve computational tractability, generalized Lagrangian duality and strong duality for moment problems are employed to reformulate the model as a deterministic semidefinite programming problem with linear matrix inequality constraints.

Given the first stage decision vector x, the inner problem identifies the probability distribution in the ambiguity set that maximizes the expected operating cost. The original formulation is:

$$\Psi(x)=\max_{P\in\mathscr{D}}\mathbb{E}_P[Q(x,\xi)]=\max_{P\in\mathscr{D}}\int_{\Xi}Q(x,\xi)d\mathbb{P}(\xi) \tag{49}$$

where $Q(x,\xi)$ is the optimal second stage operating cost under planning decision x and uncertainty realization ξ.

Substituting the moment-based ambiguity set $\mathscr{D}$ defined in Section 3.2 and representing the probability distribution by the Dirac measure, the inner problem is reformulated as the following standard moment problem:

$$\begin{aligned}&\Psi(x)=\sup_{\mathbb{P}}\int_{\Xi}Q(x,\xi)d\mathbb{P}(\xi)\\ &s.t.\begin{cases}\int_{\Xi}d\mathbb{P}(\xi)=1 & :\lambda\\ \int_{\Xi}\xi d\mathbb{P}(\xi)=\mu_0 & :\nu\\ \int_{\Xi}(\xi-\mu_0)(\xi-\mu_0)^{\top}d\mathbb{P}(\xi)\leq\gamma_1\Sigma_0 & :\Gamma\\ \mathbb{P}(\xi)\geq 0,\forall\xi\in\Xi\end{cases}\end{aligned} \tag{50}$$

where λ is a scalar, $\nu\in\mathbb{R}^{N_\xi}$ is a vector, and $\Gamma\in\mathbb{S}_{+}^{N_\xi}$ is a positive semidefinite matrix. They are the dual variables associated with the probability normalization constraint, the first moment constraint, and the second moment constraint, respectively.

The corresponding Lagrangian function $\mathbb{P}$ with respect to the primal variable λ,ν,Γ and the dual variables $\mathscr{L}(\mathbb{P},\lambda,\nu,\Gamma)$ is constructed:

$$\begin{aligned}\mathscr{L}=&\int_{\Xi}Q(x,\xi)d\mathbb{P}(\xi)-\lambda\left[\int_{\Xi}d\mathbb{P}(\xi)-1\right]-\nu^{\top}\left(\int_{\Xi}\xi d\mathbb{P}(\xi)-\mu_0\right)\\ &-Tr\left\{\Gamma\left[\int_{\Xi}(\xi-\mu_0)(\xi-\mu_0)^{\top}d\mathbb{P}(\xi)-\gamma_1\Sigma_0\right]\right\}\end{aligned} \tag{51}$$

After collecting like terms and combining the integral terms associated with the probability distribution $\mathbb{P}$, the Lagrangian becomes:

$$L=\int_{\Xi}[Q(x,\xi)-\lambda-\nu^{\top}\xi-(\xi-\mu_0)^{\top}\Gamma(\xi-\mu_0)]d\mathbb{P}(\xi)+\lambda+\nu^{\top}\mu_0+\gamma_1 Tr(\Gamma\Sigma_0) \tag{52}$$

According to strong duality, the optimal value of the primal maximization problem equals that of its dual minimization problem. To guarantee a bounded dual function, the integrand must be nonpositive for every uncertainty realization $\xi\in\Xi$. Therefore, the inner maximization problem is equivalently transformed into the following dual minimization problem:

$$\begin{aligned}&\min_{\lambda,\nu,\Gamma}\lambda+\nu^{\top}\mu_0+\gamma_1 Tr(\Gamma\Sigma_0)\\ &s.t.\begin{cases}\lambda+\nu^{\top}\xi+(\xi-\mu_0)^{\top}\Gamma(\xi-\mu_0)\geq Q(x,\xi),\forall\xi\in\Xi\\ \Gamma\geq 0\end{cases}\end{aligned} \tag{53}$$

This reformulation eliminates the infinite dimensional optimization over probability distributions and converts the original two-stage distributionally robust model into the following single level deterministic optimization problem:

$$\begin{aligned}&\min_{x,\lambda,\nu,\Gamma}C_{plan}(x)+\lambda+\nu^{\top}\mu_0+\gamma_1 Tr(\Gamma\Sigma_0)\\ &s.t.\begin{cases}x\in\mathscr{X}\\ \Gamma\geq 0\\ \lambda+\nu^{\top}\xi+\xi^{\top}\Gamma\xi-2\mu_0^{\top}\Gamma\xi+\mu_0^{\top}\Gamma\mu_0\geq Q(x,\xi),\forall\xi\in\Xi\end{cases}\end{aligned} \tag{54}$$

The reformulated model still contains a semi-infinite robust constraint that must hold for every realization in the support set Ξ. Since the support set contains infinitely many uncertainty realizations, direct discretization is intractable. Therefore, the C&CG algorithm is adopted to iteratively identify the most critical scenarios and progressively approximate the original robust constraint, providing the basis for the solution procedure presented in Section 4.

4. Model solution strategy based on C&CG

The proposed two-stage distributionally robust planning model follows a two-stage decision structure but is mathematically formulated as a nested Min-Max-Min Semi-Infinite Programming (SIP) problem. The main challenge is that finite dimensional decision variables must satisfy robust constraints defined over an infinite uncertainty support set. To solve this problem efficiently, the C&CG algorithm is adopted to reformulate the original model into a bi-level iterative framework. Instead of enumerating the entire uncertainty set, the algorithm iteratively constructs a finite set of representative worst-case scenarios $\mathscr{O}\subset\Xi$. In each iteration, the Master Problem (MP) optimizes first stage decisions and dual variables over the current scenario set to obtain a lower bound. The Sub-Problem (SP) then searches for the worst-case scenario that maximizes the violation of the current robust constraints. The identified scenario is added to the MP until convergence is achieved.

4.1. Overall framework of the algorithm

Based on the C&CG framework, the original model is decomposed into the MP and the SP, which are solved alternately. The MP is a relaxed Semi-Definite Programming (SDP) model defined over the current worst-case scenario set $\mathscr{O}=\{\xi_1,\ldots,\xi_k\}$. It optimizes the first stage capacity variables x and the distributionally robust dual variables (λ,ν,Γ), and its objective value provides the lower bound of the original problem. The SP is solved based on the optimal MP solution $(x^*,\lambda^*,\nu^*,\Gamma^*)$. It searches for the complete uncertainty support set Ξ for a new worst-case scenario ξ_{new} that maximizes the gap between the actual operating cost and the current robust approximation. If the maximum violation exceeds

the prescribed tolerance, the upper bound is updated, the new scenario is added to $\mathscr{O}$, and the corresponding variables and constraints are generated for the next MP iteration.

4.2. Construction of the master problem

The MP minimizes the investment cost and the worst-case expected operating cost. Based on the dual reformulation derived in Section 3.4, auxiliary operating variables $\mathscr{O}$ are introduced for the finite scenario set y_k. The original semi-infinite constraint is therefore relaxed to the current scenario set $\xi \in \mathscr{O}$ generated in the current iteration. Thus, the MP is formulated as the following SDP model:

$$\min_{x,\lambda,\nu,\Gamma,\{y_k\}} C_{plan}(x) + \lambda + \nu^\top \mu_0 + \gamma_1 Tr(\Gamma\Sigma_0)$$
$$\forall \xi_k \in \mathscr{O}, s.t. \begin{cases} x \in \mathscr{X} \\ \Gamma \geq 0 \\ \lambda + \nu^\top \xi_k + (\xi_k - \mu_0)^\top \Gamma(\xi_k - \mu_0) \geq C_{op}(y_k) + \phi \cdot \Delta P_{net}(y_k) \\ Gx + Hy_k \geq h(\xi_k) \end{cases} \tag{55}$$

The first two terms of the objective function represent the investment cost and fixed operation and maintenance cost. The remaining terms describe the upper bound of the worst-case expected operating cost through the dual variables. The third constraint is the robust cut, where the left hand side denotes the moment-based robust cost bound and the right hand side represents the operating cost under the corresponding scenario. The fourth constraint guarantees feasible operation y_k for every generated worst-case scenario ξ_k while satisfying power balance, ramping, and other physical constraints.

4.3. Construction of the sub-problem

The SP verifies whether the current MP solution satisfies robustness over the entire uncertainty set. Specifically, it searches for an unidentified scenario whose operating cost exceeds the current robust upper bound $\mathscr{O}$ that results in an operational cost significantly exceeding the upper bound set by the current dual variables. Given the optimal MP solution $x^*, \lambda^*, \nu^*, \Gamma^*$, the violation function $V(\xi)$ is defined, leading to the following optimization problem:

$$V(\xi) = Q(x^*, \xi) - [\lambda^* + \nu^{*\top}\xi + (\xi - \mu_0)^\top \Gamma^* (\xi - \mu_0)] \tag{56}$$

$$\max_{\xi \in \Xi} V(\xi) = \max_{\xi \in \Xi} \left\{ \min_y (c^\top y) - \left[\lambda^* + \nu^{*\top}\xi + \xi^\top \Gamma^* \xi - 2\mu_0^\top \Gamma^* \xi + \mu_0^\top \Gamma^* \mu_0\right] \right\}$$
$$s.t. Hy \geq h(\xi) - Gx^* \tag{57}$$

where $c^\top y$ is the operating cost including penalty terms under scenario ξ. The SP aims to find the scenario $V(\xi)$ that maximizes ξ^*; If the maximum violation exceeds the convergence tolerance $\max V(\xi^*) > \varepsilon$, the current solution is regarded as non-robust.

The SP is formulated as a Max-Min bi-level optimization problem. The outer level searches the uncertainty support set Ξ for the worst-case scenario ξ^*, while the inner level minimizes the operating cost under the given scenario ξ and capacity configuration x^*. Since the inner problem is a Linear Programming (LP) problem that satisfies the Slater condition, strong duality holds. Therefore, the inner optimization is replaced by its KKT conditions.

Let π denote the dual variables of the inner problem. The resulting reformulation consists of primal feasibility, dual feasibility, complementary slackness, and strong duality constraints, as follows:

$$\begin{cases} Hy \geq h(\xi) - Gx^* \\ H^\top \pi = c, \pi \geq 0 \\ \pi^\top [Hy - h(\xi) + Gx^*] = 0 \\ c^\top y = \pi^\top [h(\xi) - Gx^*] \end{cases} \tag{58}$$

The strong duality equality is substituted into the outer objective to eliminate the inner variables y. The complementary slackness conditions are linearized using the Big M method. The SP is finally reformulated as the following single level MISOCP model:

$$\max_{\xi,\pi,u} \pi^\top [h(\xi) - Gx^*] - [\xi^\top \Gamma^* \xi + (\nu^* - 2\Gamma^* \mu_0)^\top \xi] - \left(\lambda^* + \mu_0^\top \Gamma^* \mu_0\right)$$
$$s.t. \begin{cases} \xi_{\min} \leq \xi \leq \xi_{\max}, |\xi - \mu_0| \leq \gamma_2 \sigma \\ H^\top \pi = c \\ 0 \leq \pi \leq M \cdot u \\ 0 \leq Hy - h(\xi) + Gx^* \leq M \cdot (1 - u) \\ u \in \{0,1\}^{N_{con}} \end{cases} \tag{59}$$

Since $h(\xi)$ mainly consists of linear functions of the uncertain variables ξ, the resulting objective is dominated by convex quadratic terms. The reformulated model is convex and can be solved efficiently using commercial solvers such as Gurobi.

4.4. Algorithm flow and convergence criteria

The proposed solution procedure alternates between the MP and the SP until convergence. At initialization, the convergence tolerance ε, maximum iteration number, and initial worst-case scenario set $\mathscr{O}$ are specified. The initial scenario set is generated by K-means clustering of historical renewable generation and computing workload samples. The upper and lower bounds are initialized simultaneously.

During each iteration, the MP is first solved over the current scenario set to obtain the optimal investment decisions and dual variables, and the lower bound is updated. The SP is then solved to identify the current worst-case scenario ξ_{new} and the corresponding maximum violation V_{max}, which is used to update the upper bound. If the maximum violation satisfies $V_{max} \leq \varepsilon$, the algorithm terminates because all robust con-

Table 1
Algorithm logic pseudocode.

Algorithm 1: C&CG Algorithm for Distributionally Robust Planning
Input: System parameters, moment information (μ_0, Σ_0), robust parameters (γ_1, γ_2), tolerance ε.
Output: Optimal capacity configuration x^*.
1 Initialize: $LB = -\infty$, $UB = +\infty$, iteration counter $k = 1$, worst-case scenario set $\mathscr{O} = \varnothing$.
2 While $Gap > \varepsilon$ **do**
3 Step 1: Solve the Master Problem (MP)
Solve the relaxed SDP model over set $\mathscr{O}$:
$MP: \min_{x,\lambda,\nu,\Gamma,\{y_k\}} C_{plan}(x) + \lambda + \nu^\top \mu_0 + \gamma_1 Tr(\Gamma\Sigma_0)$
$\forall \xi_k \in \mathscr{O}, s.t. \begin{cases} \lambda + \nu^\top \xi_k + (\xi_k - \mu_0)^\top \Gamma(\xi_k - \mu_0) \geq C_{op}(y_k) + \phi \cdot \Delta P_{net}(y_k) \\ Gx + Hy_k \geq h(\xi_k) \end{cases}$
Obtain optimal solution $\{x^*, \lambda^*, \nu^*, \Gamma^*\}$.
Update Lower Bound: $LB = Objective^*_{MP}$.
4 Step 2: Solve the Sub-Problem (SP)
Fix $x^*, \lambda^*, \nu^*, \Gamma^*$ and solve the single-level MISOCP (reformulated via KKT):
$SP: V_{\max}(\xi) = \max_{\xi \in \Xi} \{Q(x^*, \xi) - [\lambda^* + \nu^{*\top}\xi + (\xi - \mu_0)^\top \Gamma^* (\xi - \mu_0)]\}$
Obtain the worst-case scenario ξ_{new} and maximum violation $V_{\max}$.
Update Gap: $Gap = V_{max}$. (Or update UB and Gap based on specific logic)
5 Step 3: Convergence Check and Generation
If $Gap \leq \varepsilon$ **then**
Break (Algorithm Converged).
Else
Update scenario set: $\mathscr{O}_{new} = \mathscr{O} \cup \{\xi_{new}\}$.
Generate new operation variables y_{new} and constraints for MP.
$k = k + 1$.
End If
6 End While
7 Return optimal solution x^*.

straints are satisfied. Otherwise, the identified worst-case scenario ξ_{new} is added to the scenario set $\mathscr{O}$, new operating variables and constraints are generated, and the next iteration begins. The complete procedure is summarized in Table 1.

5. Case analysis

5.1. Case settings

To verify the effectiveness and applicability of the proposed distributionally robust planning model for computing-electricity coupling in a DCC under supply-demand uncertainties, a DCC demonstration project located in northern China is selected as the case study. The planning horizon is 30 years, and the discount rate is 8%. According to workload response characteristics, the computing tasks are classified into FL, TSL, and STL to represent different workload types. Historical wind speed and photovoltaic irradiance data from the region are used for renewable generation modeling. Considering the strong volatility and seasonal variation of renewable generation and computing workloads, K-means clustering is applied to the 8760 hourly historical samples to extract four typical days representing spring, summer, autumn, and winter. The time resolution of each typical day is 1 h. The renewable energy output curves and data center load profiles are provided in Appendix Fig. 1 and Appendix Fig. 2.

To characterize supply-demand uncertainties and their correlations, a data-driven uncertainty modeling framework is adopted. An LSTM network is used to forecast wind power, photovoltaic generation, and baseline computing workloads, yielding the predicted mean vector. The covariance matrix is estimated from historical forecasting errors [42]. Based on the obtained moment information, a distributionally robust ambiguity set is constructed using first order moments, second order moments, and the support set of uncertain variables. The confidence levels of the mean vector and covariance matrix are both set to 95% to account for distributional deviations caused by forecasting errors and limited historical samples. The proposed model is implemented in Python 3.13 and solved using Gurobi 12.0.0. The relative optimality gap is set to $\varepsilon = 10^{-4}$. The investment costs, operation and maintenance costs, and technical parameters of the CCHP units, energy storage systems, and server equipment are provided in Appendix Table 1 and Appendix Table 2.

Table 2
Scenario settings.

	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
CCHP System	×	✓	✓	✓	✓
Load Scheduling Mechanism	✓	×	✓	✓	✓
SP	×	×	✓	×	×
RO	×	×	×	✓	×
DRO	✓	✓	×	×	✓

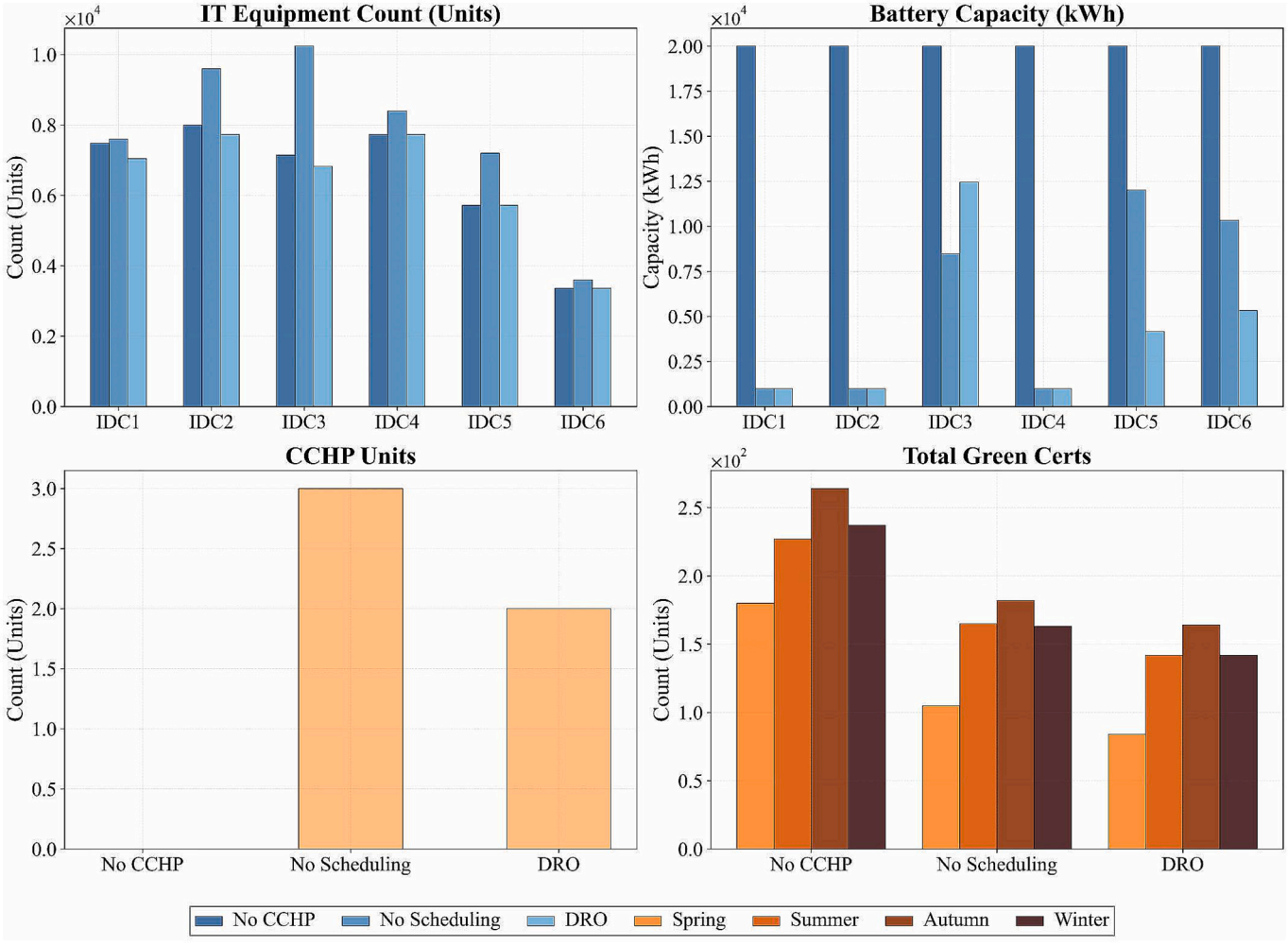

Fig. 2. Configuration results at the computing-power and electricity coupling mechanism level.

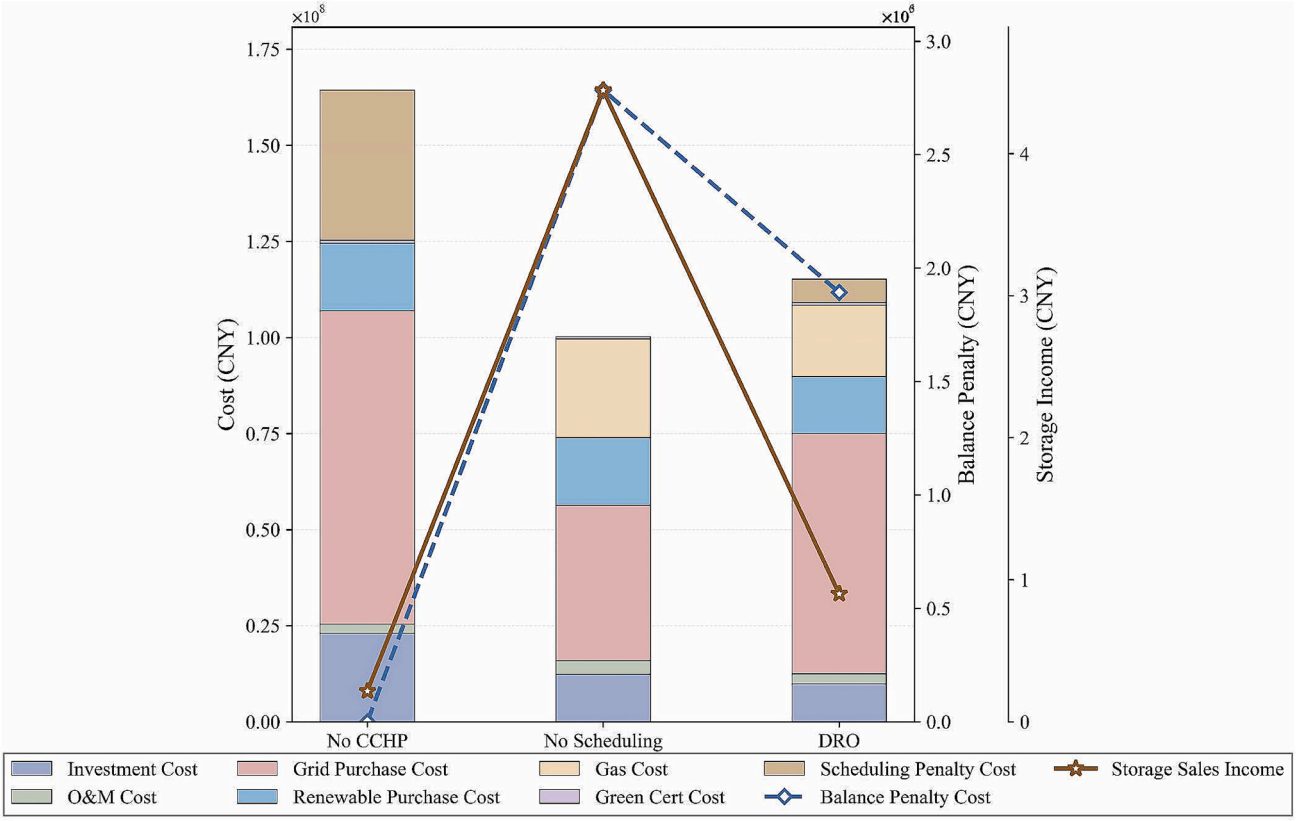

Fig. 3. Cost and benefit results at the computing-power and electricity coupling mechanism level.

5.2. Capacity configuration and multi-dimensional economic benefit verification

To evaluate the effects of computing-electricity coupling and uncertainty modeling, five comparison scenarios are designed. The scenarios isolate the impacts of energy infrastructure, workload flexibility, and optimization methods through controlled variable analysis. Capacity configuration, economic performance, and multi energy operation are compared. The scenario settings are summarized in Table 2.

5.2.1. Impact of computing-electricity coupling strategies

The computing-electricity coupling strategy significantly affects the optimal capacity configuration of the DCC. The configuration results are presented in Fig. 2. In Scenario 1, without the CCHP system, the system loses dispatchable generation and combined cooling heating and power capability. Consequently, all DCs require the maximum energy storage capacity to maintain multi energy balance. In Scenario 2, without workload flexibility, the required number of IT servers increases substantially because each DC must be sized according to its local peak computing demand.

Workload flexibility enables computing tasks to be shifted across time and space, providing a virtual energy storage effect and reducing physical capacity redundancy. As a result, the required capacities of IT servers and energy storage systems are both reduced. Economically, the scenario without the CCHP system incurs the highest lifecycle cost because it relies heavily on electricity purchases during periods of low renewable generation.

Fig. 3 compares the cost components of different scenarios. Without workload flexibility, electricity purchase costs decrease slightly, but renewable fluctuations increase net load balancing penalties and reduce energy storage arbitrage revenue. In contrast, the DRO scenario exploits demand side flexibility to reduce equipment investment and operation costs while increasing electricity sales revenue from energy storage.

Table 3
Evaluation indicators at the computing-power and electricity coupling mechanism level.

	Scenario 1	Scenario 2	Scenario 3
Annual wind curtailment rate	2.44%	2.45%	3.17%
Annual photovoltaic curtailment rate	–	0.16%	0.21%
QoS index	92.5%	98.5%	99.2%
Equipment utilization efficiency	75.2%	62.3%	86.5%

Table 3 compares the operational performance of different computing-electricity coupling strategies.

The DRO scenario achieves the highest equipment utilization of 86.5% while maintaining a QoS of 99.2%. By comparison, Scenario 2 maintains acceptable QoS but its equipment utilization decreases to 62.3%, indicating substantial idle capacity. The annual photovoltaic curtailment rate is also limited to 0.21%, demonstrating effective renewable accommodation. These results confirm that workload flexibility improves renewable utilization and resource allocation with only a small scheduling cost.

5.2.2. Comparative analysis of uncertainty processing method scenarios

Different uncertainty modeling methods lead to different investment decisions. Fig. 4 compares the corresponding capacity configurations. The traditional two-stage RO model adopts the most conservative strategy by assuming the worst-case of zero renewable generation and peak demand. Consequently, larger CCHP capacities and maximum energy storage capacities are installed. In contrast, the DRO model excludes extremely unlikely scenarios through the ambiguity set, resulting in more economical capacity configurations.

Fig. 5 compares the lifecycle cost components. SP minimizes investment cost based on historical distributions but suffers the highest balancing penalty under out of sample conditions. RO significantly

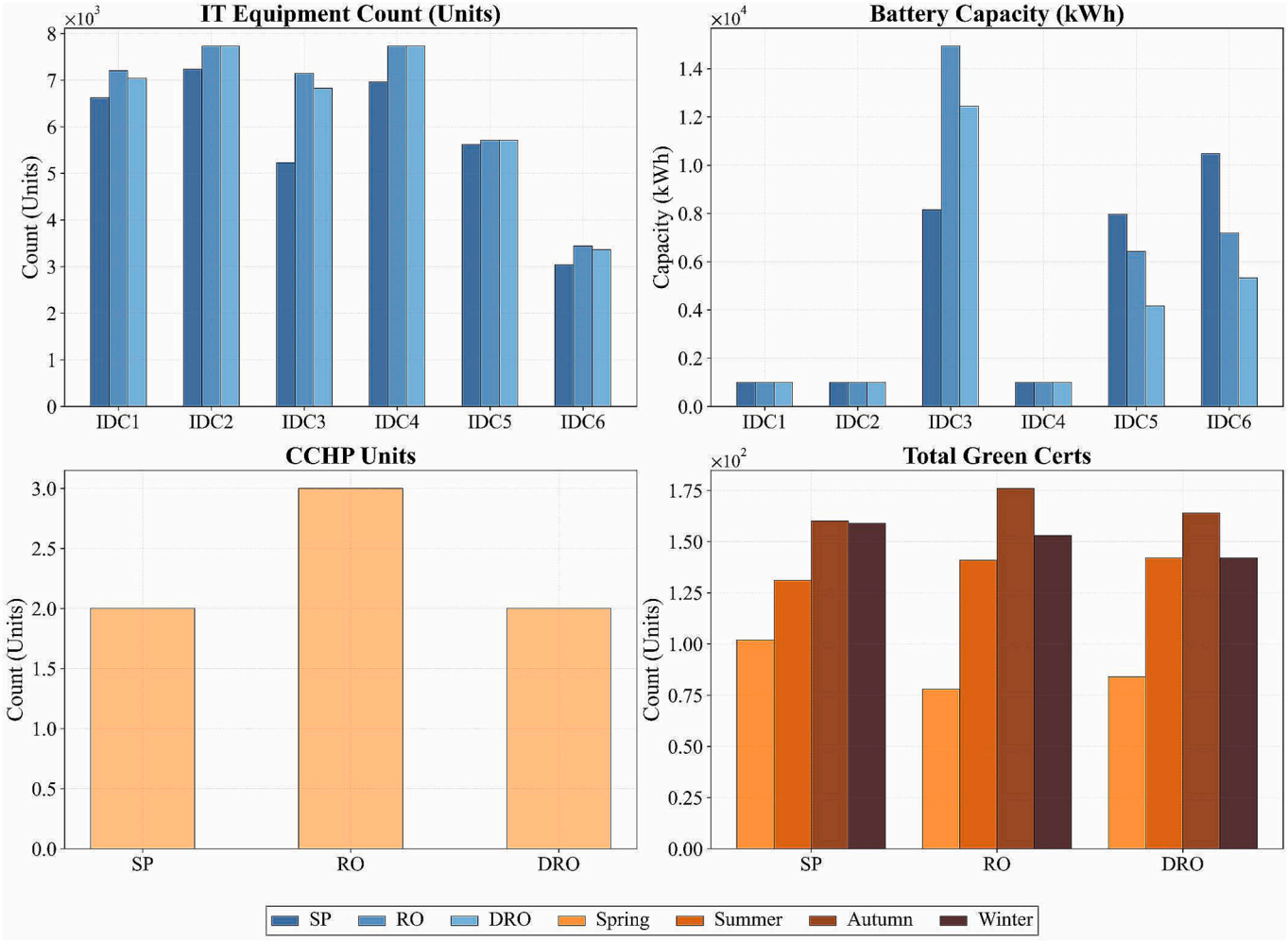

Fig. 4. Configuration results at the uncertainty optimization method level.

increases investment and green certificate costs because of excessive capacity redundancy. DRO achieves a better balance between economy and robustness through moderate reserve capacity and adaptive renewable scheduling.

Table 4 summarizes the operational and computational performance of different uncertainty modeling methods.

The proposed DRO C&CG framework converges within 8 iterations, reaching a relative optimality gap of 0.01% after approximately 850 s. Compared with the single level SP model, the solution time increases because the ambiguity set is explicitly considered. Nevertheless, the MP and SP converge smoothly, avoiding exhaustive exploration of the infinite dimensional distribution space.

The DRO model maintains a QoS of 99.2% while achieving 86.5% equipment utilization. By contrast, RO guarantees robustness at the expense of excessive capacity redundancy, reducing equipment utilization to 55.6%. These results demonstrate that the proposed two-stage DRO framework solved by C&CG achieves a better balance between planning economy and robustness under supply-demand uncertainties, while maintaining high QoS and efficient resource utilization in the DCC.

5.3. *Synergistic operation of multi energy flows and computing power*

This section analyzes the operating characteristics of the DCC under the optimal planning scheme. The analysis focuses on multi energy flow coordination, energy storage operation, spatiotemporal scheduling of computing workloads, and the role of computing-electricity coupling in improving supply-demand coordination and renewable energy utilization.

5.3.1. *Power balance analysis under multi source complementarity*

The power balance on typical days in different seasons demonstrates the capability of the multi energy system to accommodate supply-demand mismatches. The total load of the DCC exhibits pronounced daily peak and valley variations, requiring sufficient operational flexibility from the energy supply system. By coordinating wind power, photovoltaic generation, energy storage, and CCHP units, the system maintains a robust multi energy balance. The corresponding power profiles are shown in Fig. 6.

The system maintains real time power balance through coordinated dispatch of multiple energy resources. Photovoltaic generation exhibits a typical single peak profile and supplies a large share of daytime computing demand. Between 18:00 and 21:00, photovoltaic output decreases while system demand reaches its evening peak. During this period, energy storage discharges and CCHP units increase output, reducing electricity purchases from the utility grid. Gas units provide dispatchable generation, energy storage mitigates renewable fluctuations, and the utility grid supplies the remaining power imbalance. This coordinated strategy effectively compensates for renewable intermittency and ensures reliable operation of IT equipment.

5.3.2. *Analysis of energy storage operation on typical days*

Energy storage serves as the key flexible resource connecting intermittent renewable generation and time varying loads. Under the

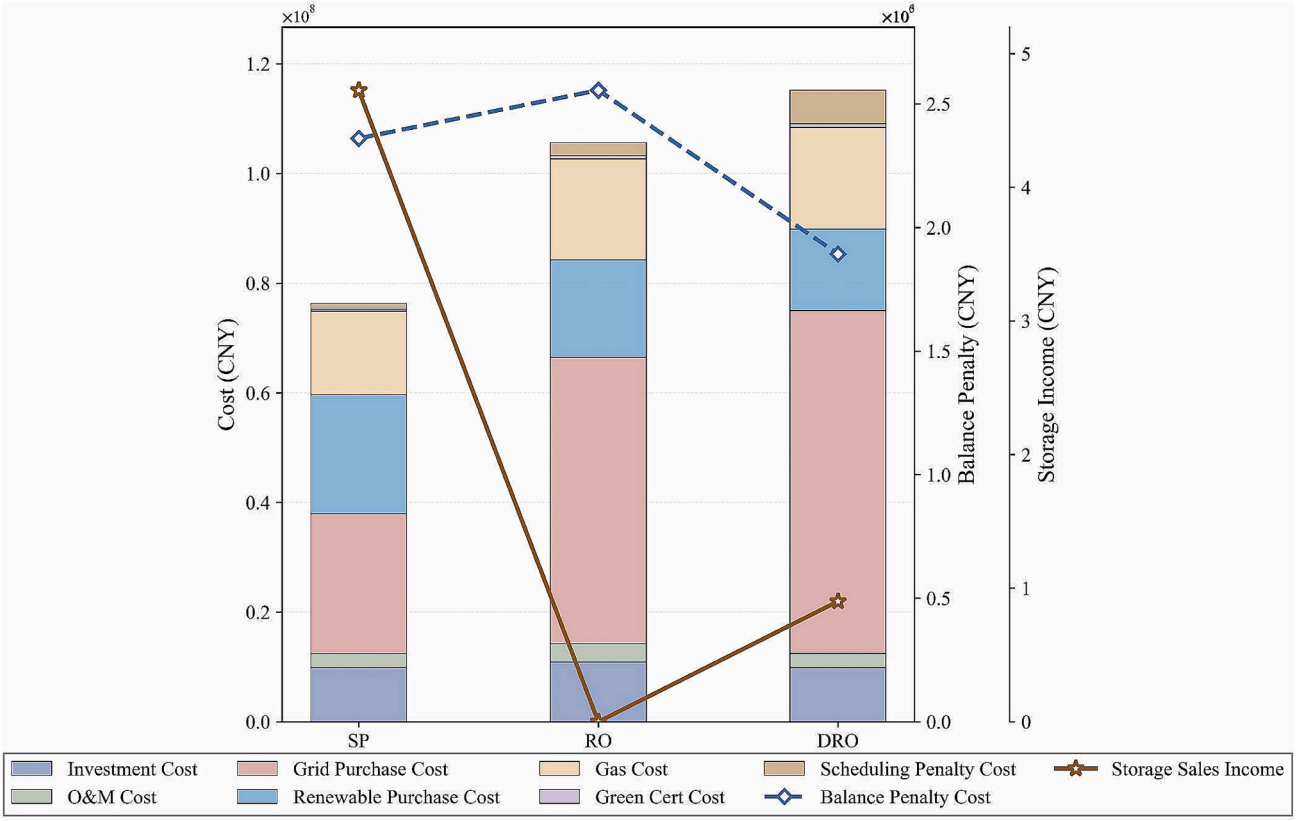

Fig. 5. Cost and benefit results at the uncertainty optimization method level.

computing-electricity coupling framework, its operation responds to renewable fluctuations while coordinating with workload scheduling. The charging and discharging profiles for typical days are presented in Fig. 7.

The energy storage system exhibits strong operational flexibility. From 1:00 to 6:00, it charges during off peak electricity price periods when wind power output is relatively stable. Between 13:00 and 16:00, surplus photovoltaic generation is absorbed through additional charging, increasing renewable energy utilization.

During the morning peak from 9:00 to 12:00 and the evening peak from 19:00 to 21:00, the energy storage system operates in discharge mode, with the maximum discharge power approaching 10 MW at some nodes. This operating pattern is jointly driven by time of use electricity prices and the net load deviation penalty in the two-stage planning model. Charging during low demand periods and discharging during peak demand periods not only realizes energy arbitrage but also smooths net load fluctuations, making energy storage the primary flexibility resource of the system.

5.3.3. *Spatiotemporal scheduling analysis of computing workloads*

Workload flexibility is the primary mechanism for supply-demand coordination in the proposed DCC. Unlike conventional loads, DC workloads can be shifted across time and migrated among geographically distributed DCs according to service delay requirements. The coordinated scheduling of TSL and STL significantly improves the load profile, as illustrated in Fig. 8.

Fig. 8 shows that TSL performs peak shaving and valley filling by exploiting workload delay tolerance. During periods of low system load and abundant wind power, more TSL tasks are executed to maximize renewable energy utilization. During the evening peak from 20:00 to 22:00, part of the TSL is deferred, reducing peak electricity demand. Meanwhile, STL dynamically reallocates workloads among different DCs through high speed communication networks.

Table 4
Evaluation indicators at the uncertainty optimization method level.

	Scenario 3	Scenario 4	Scenario 5
Annual wind curtailment rate	3.17%	2.31%	1.93%
Annual photovoltaic curtailment rate	0.21%	0.32%	0.27%
QoS index	99.2%	90.1%	99.9%
Equipment utilization efficiency	86.5%	88.4%	55.6%
Number of iterations	8	–	6
Solution time	850 s	120 s	450 s
Relative convergence gap	0.01%	0.01%	0.01%

To further quantify temporal scheduling, the virtual queue length of each DC is presented in Fig. 9.

The virtual queue length indicates that TSL is mainly accumulated between 21:00 and 23:00, when electricity prices and system load are both high. To avoid high operating costs, newly arrived tasks are temporarily stored in virtual queues, and the total backlog reaches approximately 160,000 tasks. A smaller queue also appears between 11:00 and 13:00 on a typical autumn day because of temporary photovoltaic fluctuations and local supply-demand imbalance. As wind generation increases and system load decreases during late night hours, the queued tasks are gradually released. This queue based scheduling strategy provides temporal flexibility without violating QoS constraints.

The spatial scheduling results are shown in Fig. 10.

Fig. 10 shows that STL dynamically reallocates workloads when local DCs experience heavy loading or renewable energy shortages. The maximum transferred workload reaches approximately 30 MW. Through coordinated optimization of virtual queues and computing network flows, computing workloads are transformed into virtual energy flows with spatiotemporal flexibility. This mechanism provides a virtual energy storage effect without additional physical investment, significantly improving load smoothness and demonstrating the effectiveness of computing-electricity coupling.

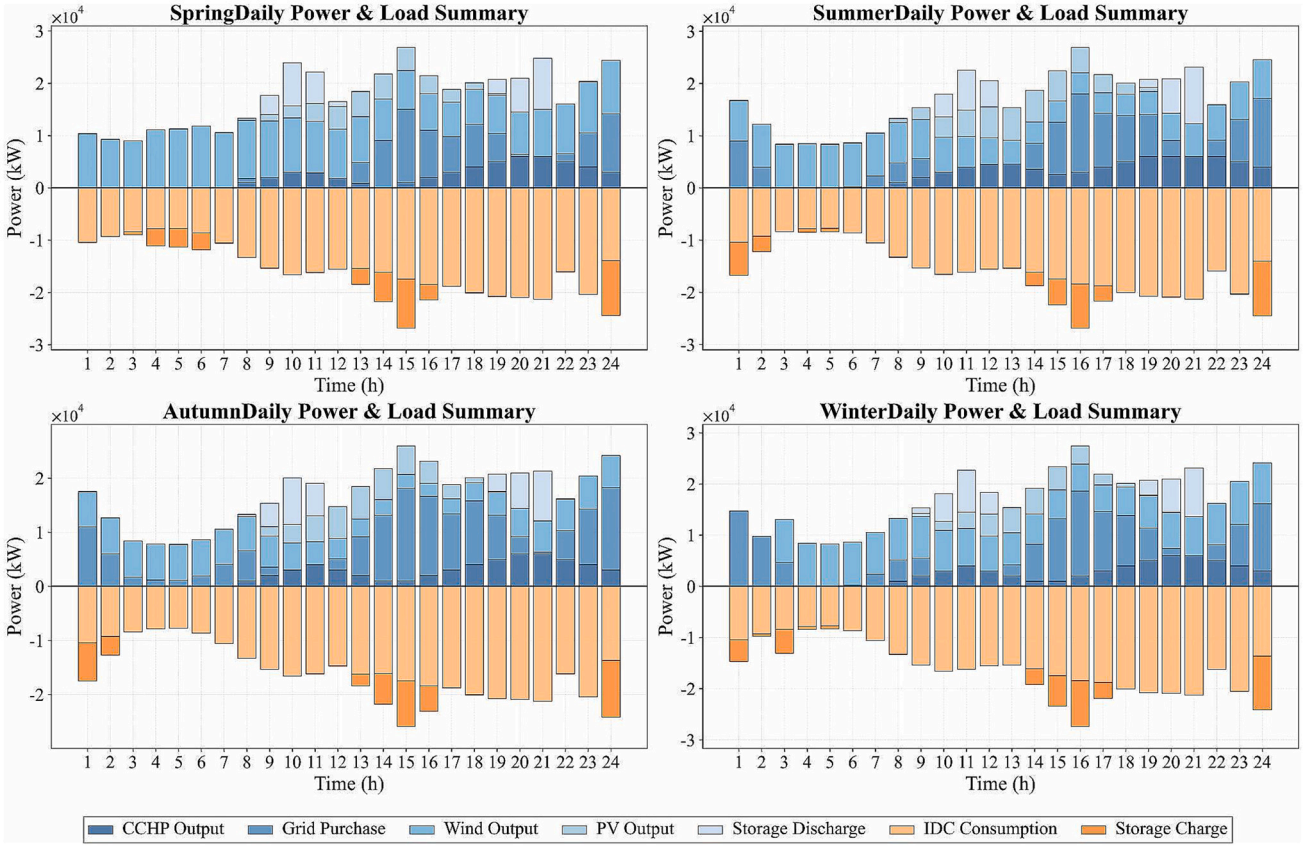

Fig. 6. Output and load under typical days of each season.

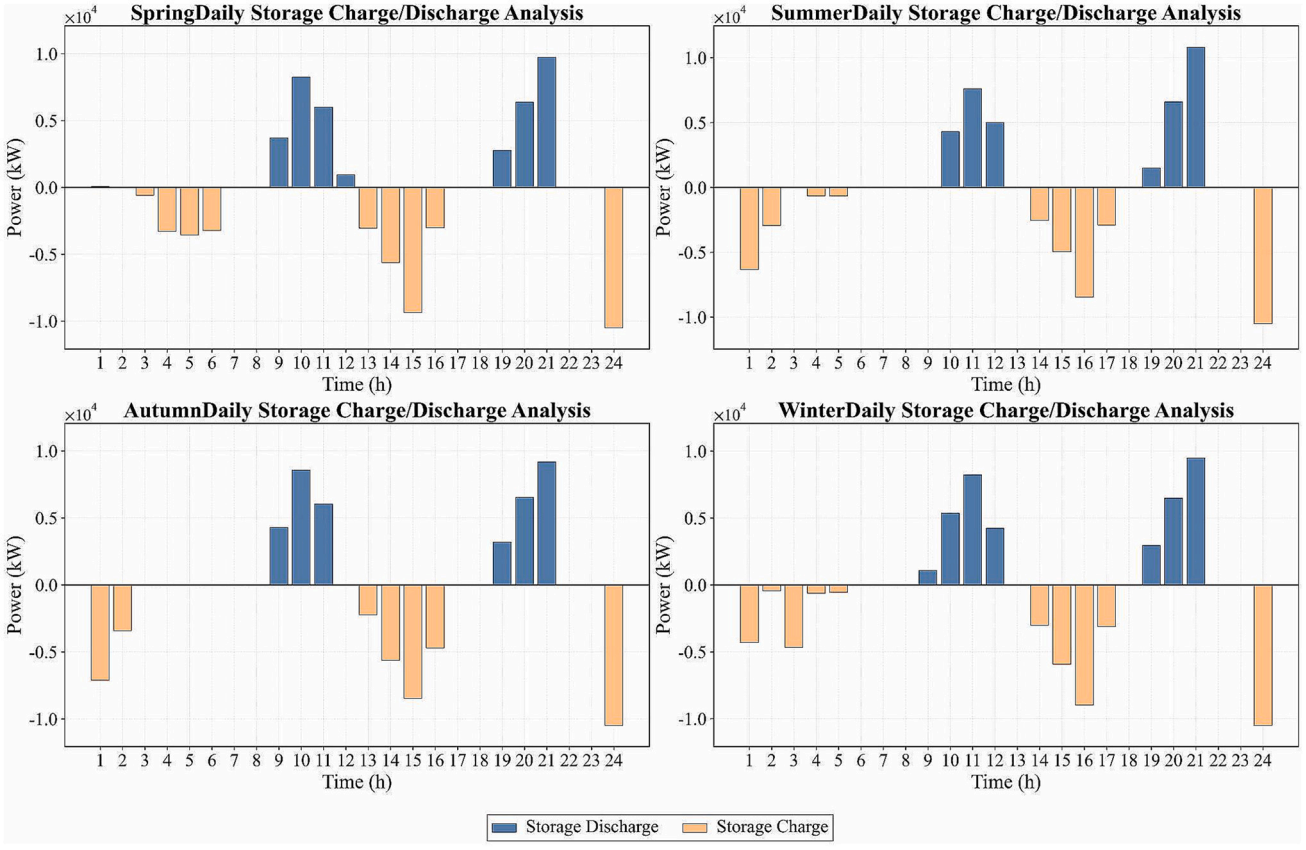

Fig. 7. Charging and discharging status of the energy storage system under typical days of each season.

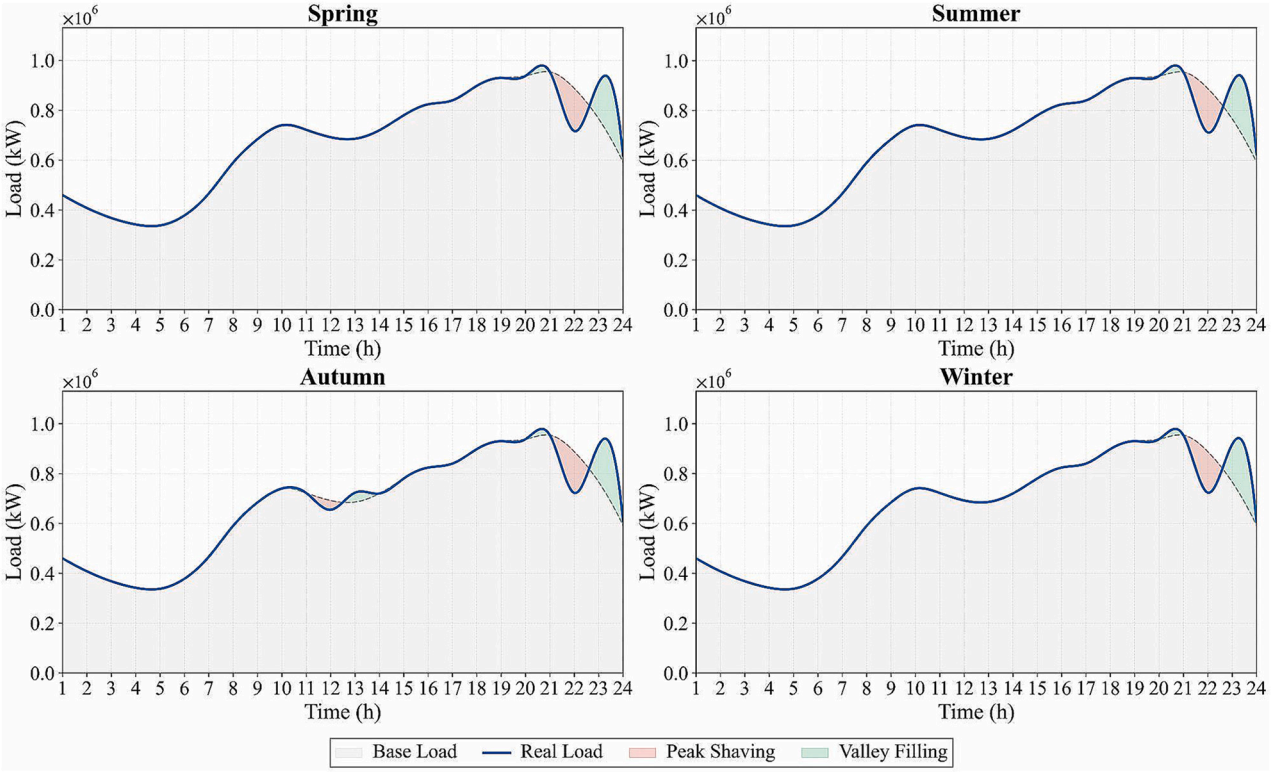

Fig. 8. Peak-shaving and valley-filling effects under typical days of each season.

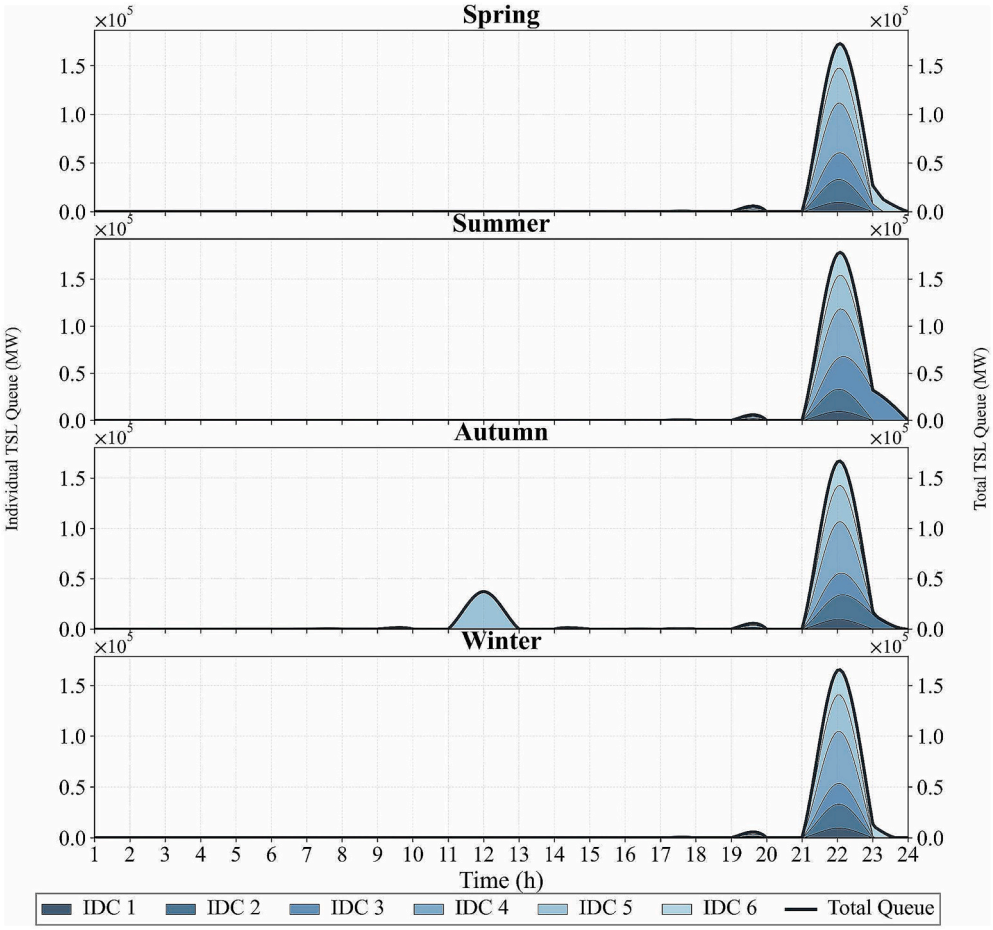

Fig. 9. Waiting queue length for time transfer of loads in each DC.

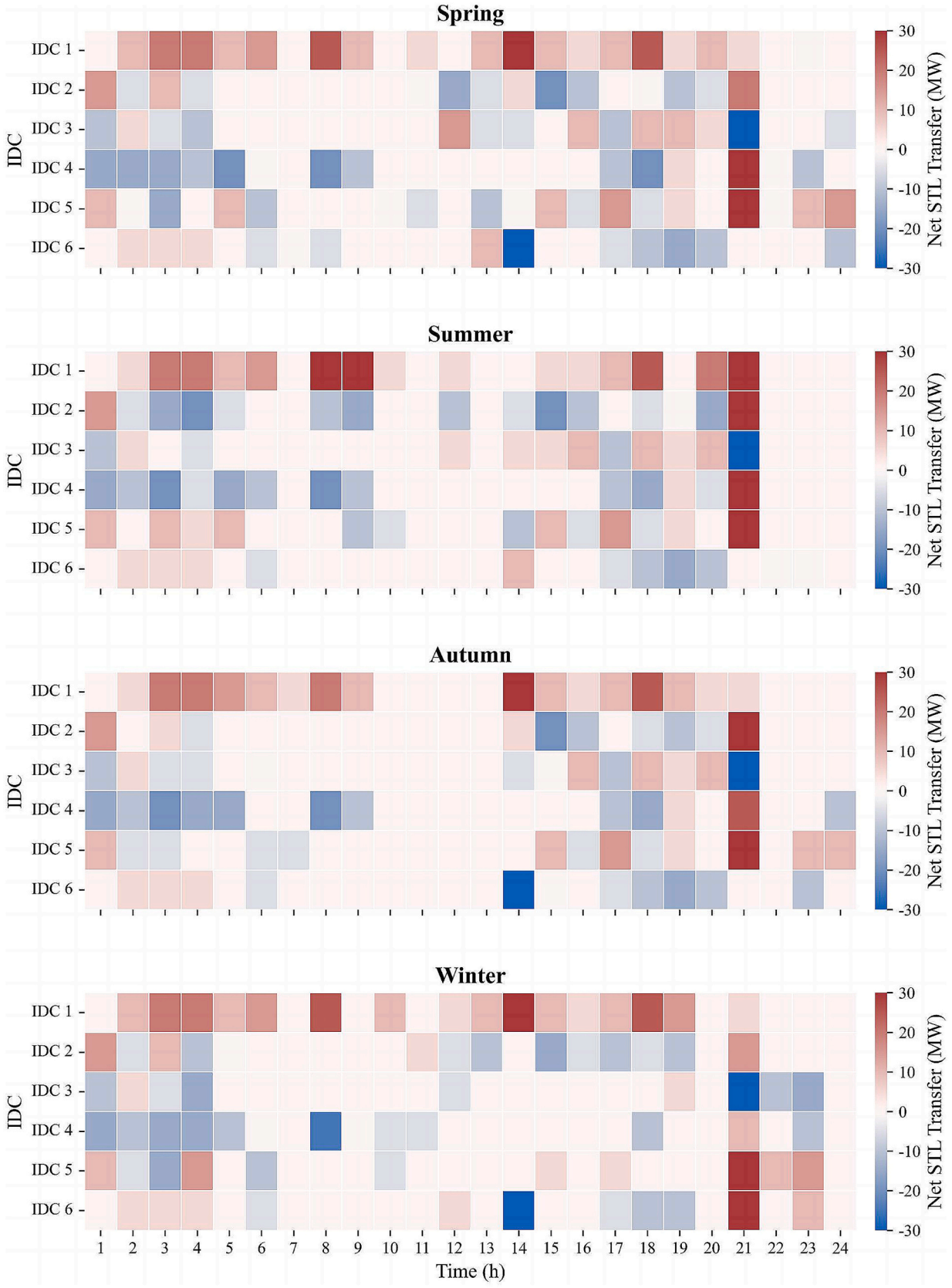

Fig. 10. Spatial scheduling results of loads in each DC.

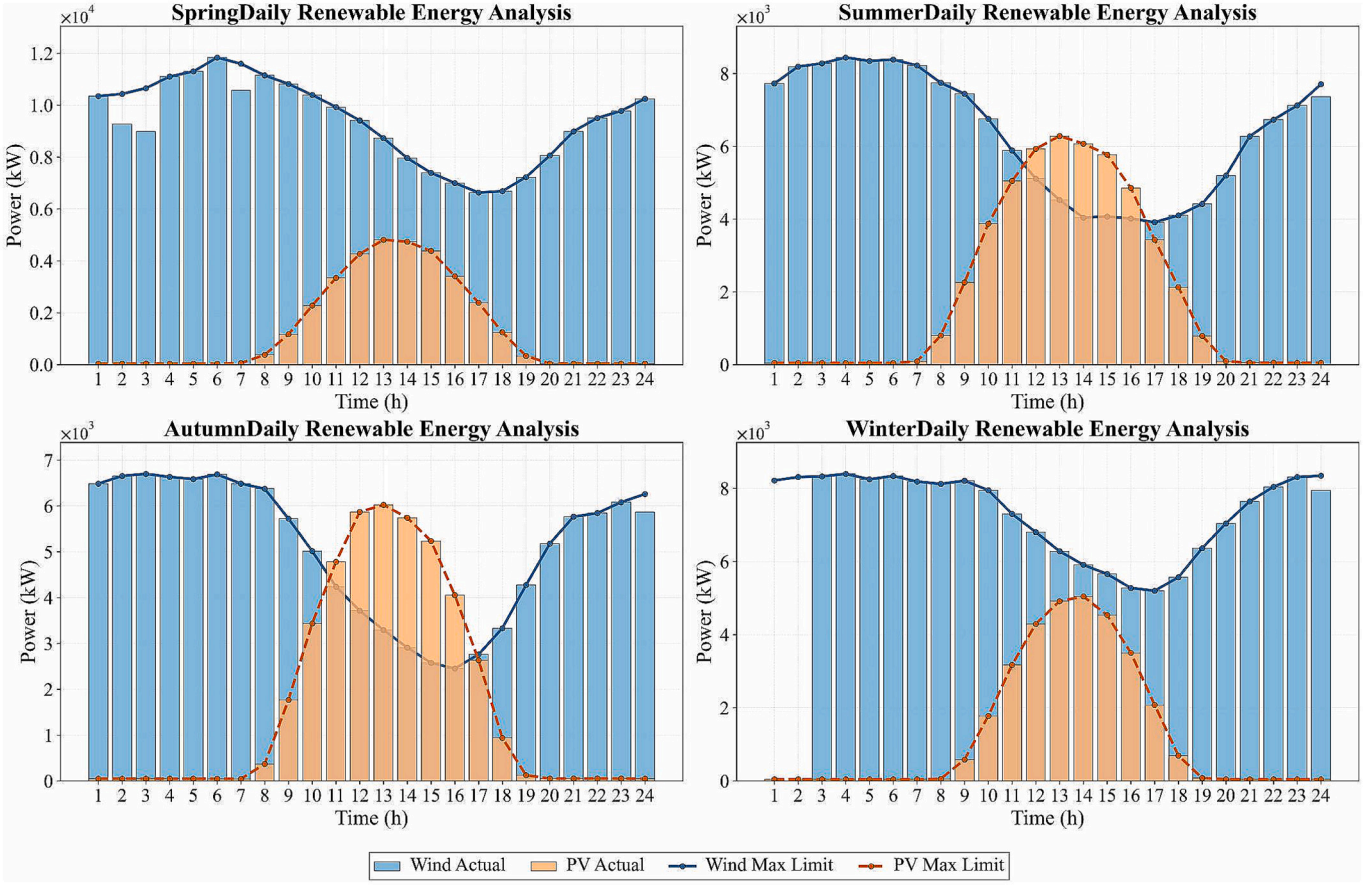

Fig. 11. Actual renewable energy output under typical days of each season.

5.3.4. *Analysis of renewable energy accommodation and supply-demand matching*

Renewable intermittency and time varying computing demand are the main factors affecting renewable energy utilization in the DCC. The coordinated operation of multiple energy resources improves renewable energy accommodation while maintaining supply-demand balance. Fig. 11 presents the theoretical and actual renewable generation under typical days.

Photovoltaic generation closely follows its theoretical upper limit during midday, indicating a high renewable utilization rate. Limited wind curtailment occurs during late night hours because stable wind generation exceeds the charging capability of energy storage and the minimum output constraints of CCHP units. A small amount of photovoltaic curtailment also appears during summer and autumn noon periods.

These results reflect the tradeoff between economy and robustness in the proposed DRO model. The Min-Max-Min framework determines the worst-case probability distribution within the ambiguity set, while the outer optimization guarantees secure operation under this distribution. Instead of pursuing complete renewable accommodation through excessive capacity expansion, the model allows limited renewable curtailment under extreme scenarios to avoid unnecessary investment in energy storage. This balance between renewable utilization and infrastructure investment enables the proposed method to achieve lower long-term planning cost while satisfying renewable energy utilization targets.

5.4. *Sensitivity analysis*

To further evaluate the robustness and adaptability of the proposed planning model under different operating conditions and policy settings, this section investigates the sensitivity of four key parameters, including the renewable energy accommodation target, DRO conservativeness, computing transmission cost, and the time-of-use electricity price. The results provide guidance for planning and operational decisions in DCC.

5.4.1. *Analysis of renewable energy accommodation weight*

The renewable energy accommodation target directly affects renewable utilization requirements and green certificate demand in DCC. To evaluate its impact, five scenarios are considered, namely conservative, robust, baseline, aggressive, and optimal.

To satisfy increasingly stringent renewable energy accommodation targets, the model expands flexible resources, particularly energy storage capacity. This trend becomes more evident under the aggressive and optimal scenarios, where several DCs require substantially larger battery capacities. Green certificate purchases also increase to compensate for the limited local renewable accommodation capability. These results indicate that higher accommodation targets increase the dependence on energy storage and other flexible resources. The corresponding cost and revenue variations are presented in Fig. 12.

Fig. 12 shows that the annual total cost first decreases and then increases as the accommodation target rises. The robust and baseline scenarios provide the best balance between investment cost, operating

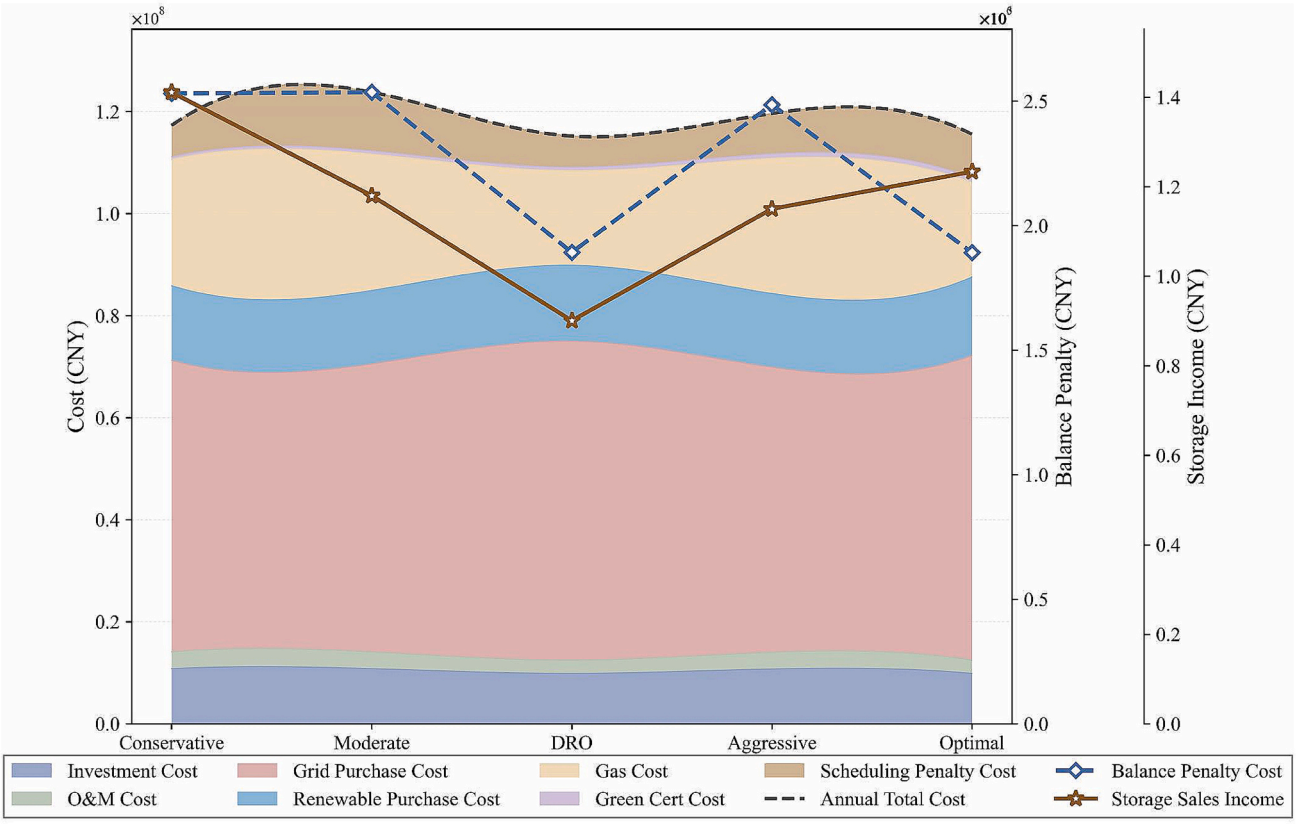

Fig. 12. Costs and revenues under different renewable energy accommodation weights.

cost, and energy storage arbitrage revenue. Further increasing the accommodation target improves renewable utilization but requires additional energy storage investment. Meanwhile, workload scheduling penalties restrict excessive workload migration, further increasing the overall system cost. Therefore, a moderate accommodation target achieves a better tradeoff between economic performance and operational reliability.

The corresponding energy flow patterns under the two extreme accommodation scenarios are illustrated in Fig. 13.

Compared with the low accommodation scenario, more wind and photovoltaic generation is routed to energy storage under the high accommodation scenario, allowing renewable energy to be shifted across time to match workload demand. Grid electricity purchases are therefore significantly reduced, resulting in lower carbon emissions. However, more frequent charging and discharging also increases storage losses. In contrast, the low accommodation scenario relies more heavily on grid electricity and CCHP generation, leading to greater renewable curtailment. These results demonstrate that renewable accommodation targets fundamentally reshape the energy flow structure of the integrated energy system.

5.4.2. *Analysis of distributionally robust conservativeness*

In the proposed DRO model, the conservativeness parameter determines the size of the ambiguity set and the level of protection against supply-demand uncertainties. To evaluate the tradeoff between robustness and economy, five conservativeness levels are examined.

As the conservativeness parameter increases, capacity planning gradually shifts toward larger reserve margins. To hedge against more severe renewable generation uncertainty and workload fluctuations within the ambiguity set, the model allocates additional dispatchable

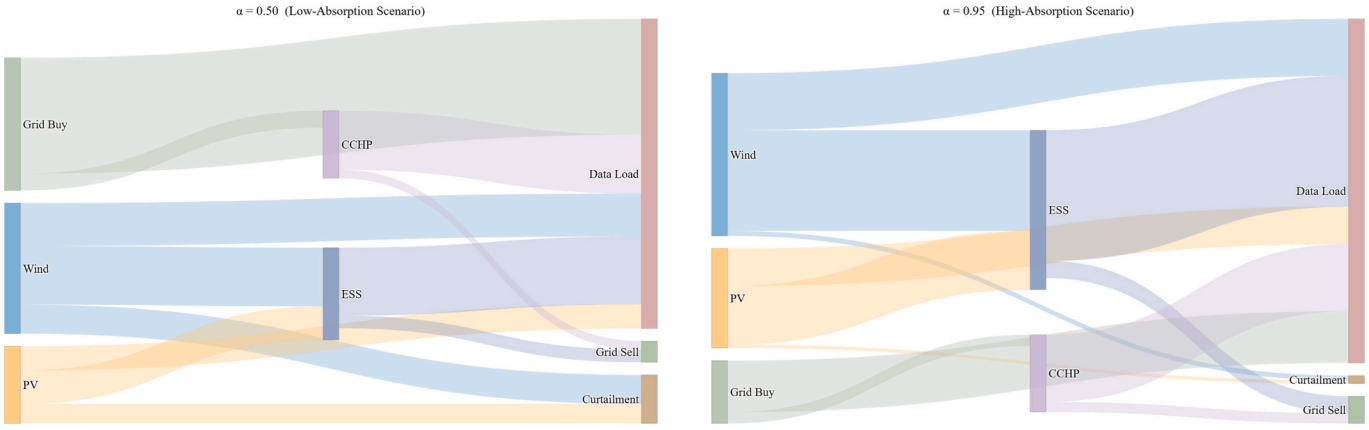

Fig. 13. Sankey diagram of energy flows under extreme renewable energy accommodation weights.

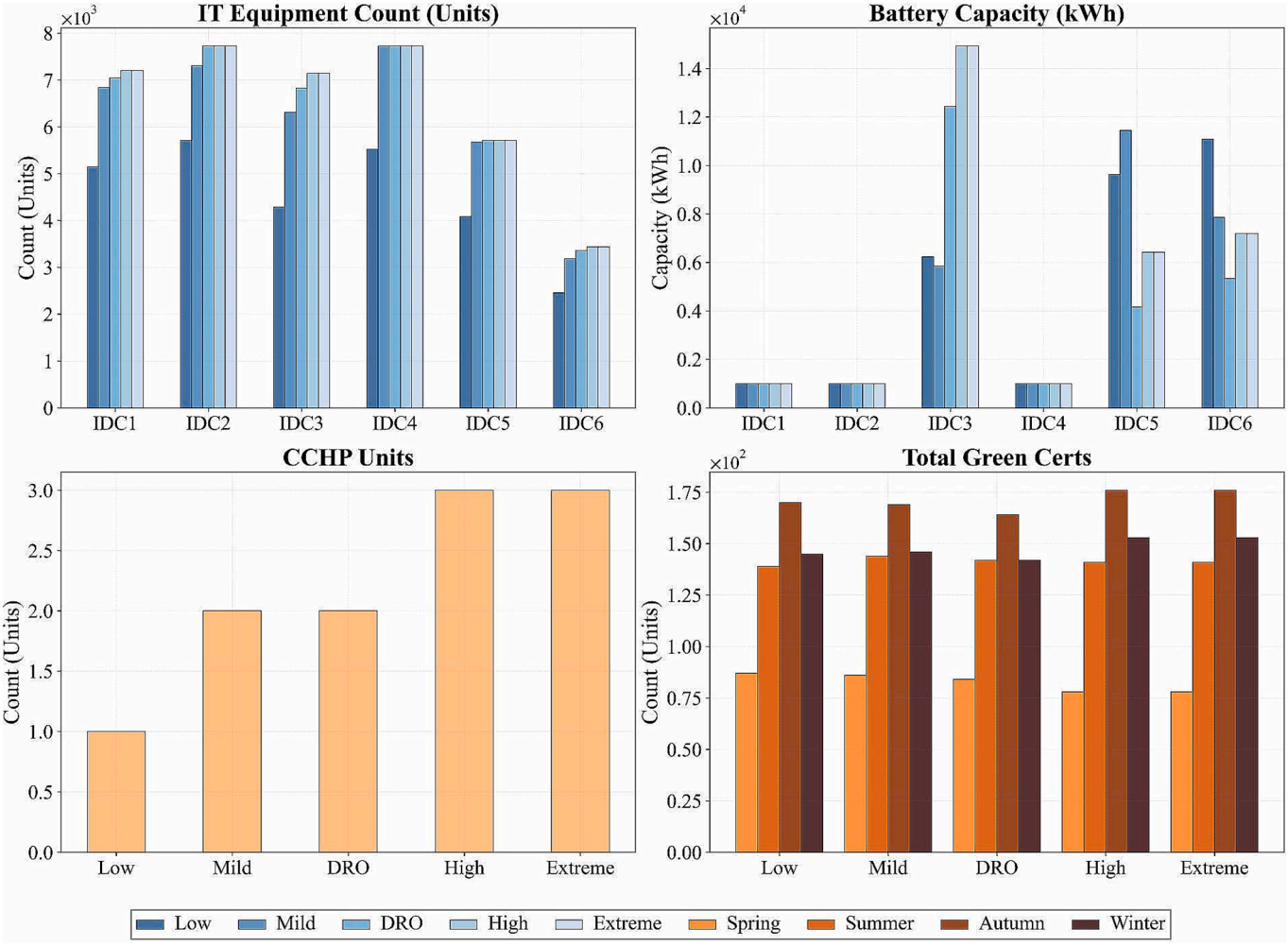

Fig. 14. Configuration results under different uncertainty conservativeness levels.

and storage resources. Specifically, the number of CCHP units increases from one under the lowest conservativeness level to three under the highest level. Energy storage capacity at several key DCs also increases accordingly. The detailed planning results are shown in Fig. 14.

Increasing reserve capacity and adopting more conservative scheduling strategies improve system resilience under adverse operating conditions. However, this improvement comes at the expense of higher planning cost. Under low conservativeness, the required reserve margin is limited and the annual total cost remains relatively low. As conservativeness increases, additional investment in reserve capacity substantially raises the total annual cost. Moreover, overly conservative planning reduces the operational flexibility of energy storage during normal periods, lowering electricity price arbitrage revenue and further increasing the net operating cost.

To further illustrate the tradeoff between economy and robustness, Fig. 15 presents the Pareto frontier of expected annual operating cost and net load fluctuation penalty.

Fig. 15 shows a clear nonlinear tradeoff between economic performance and robustness. The baseline DRO scenario lies near the Pareto optimal region, providing strong robustness with only a moderate increase in operating cost. This result suggests that a relatively low conservativeness level is appropriate for regions with stable renewable resources, whereas higher conservativeness should be adopted in regions with significant renewable variability. Selecting an appropriate conservativeness level therefore provides an effective balance between planning economy and operational risk.

5.4.3. *Analysis of cross regional transmission of computing loads*

The cross regional scheduling of STL relies on wide area network bandwidth, while network leasing and communication delay costs directly affect its economic viability. To quantify the impact of transmission costs on system planning and operation, the space transfer penalty coefficient gradually increased from 0 to 0.01. The corresponding economic performance is presented in Fig. 16.

As the cross regional transmission cost increases, the scheduling penalty first increases and then gradually stabilizes. Higher transmission costs reduce the utilization of STL, forcing the system to rely more on local physical resources to balance supply-demand uncertainties. Consequently, physical investment and grid electricity purchase costs increase, leading to a gradual increase in the total annual cost. Meanwhile, the fluctuation in energy storage revenue indicates that reduced workload flexibility requires more frequent energy storage dispatch, which limits electricity arbitrage opportunities.

Higher transmission costs also strengthen the substitution between virtual and physical flexibility resources. The corresponding configuration results are shown in Fig. 17.

When the space transfer penalty coefficient is close to zero, the model fully exploits STL as virtual energy storage. The volume of cross regional workload transfer reaches approximately 11,460 tasks, while physical energy storage capacity is limited to about 162 MWh. As the

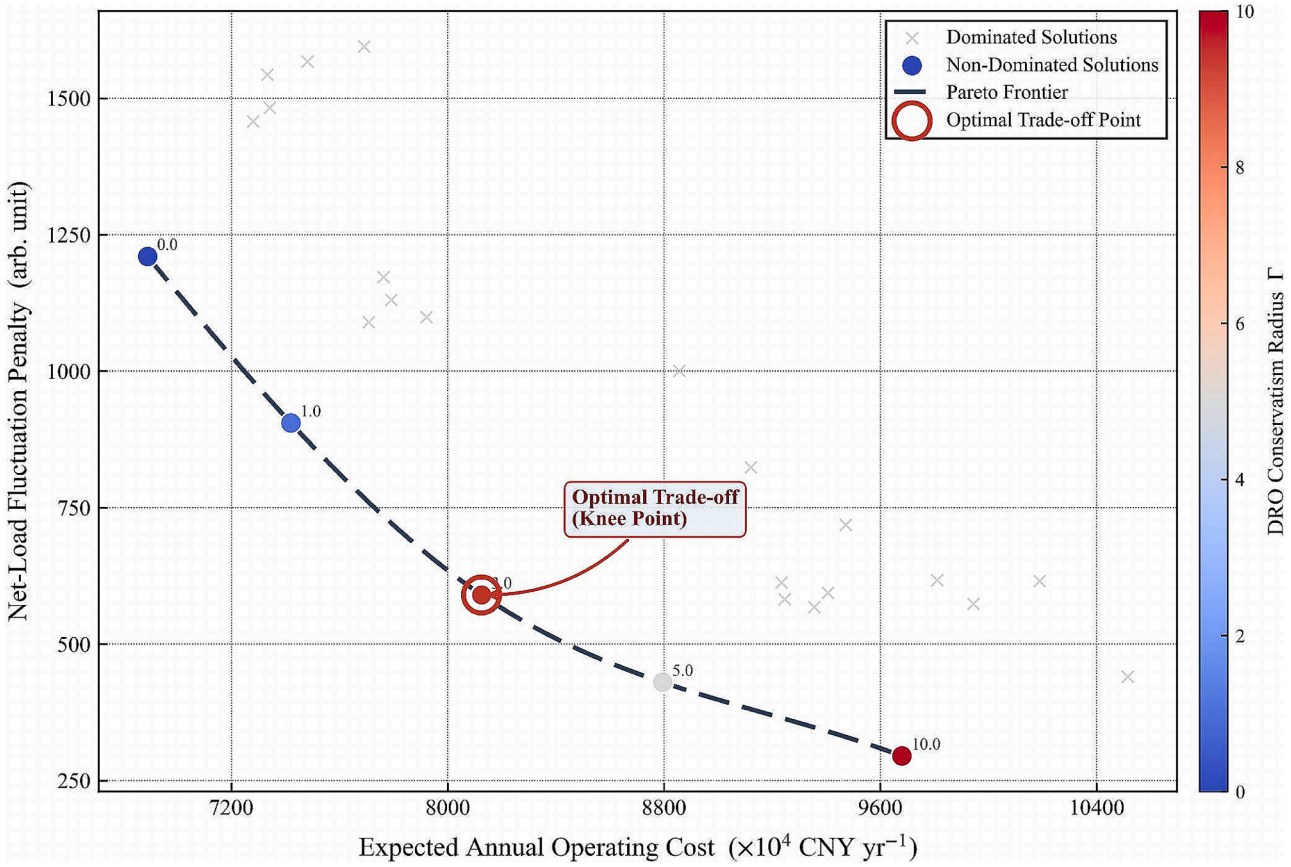

Fig. 15. Pareto frontier curve.

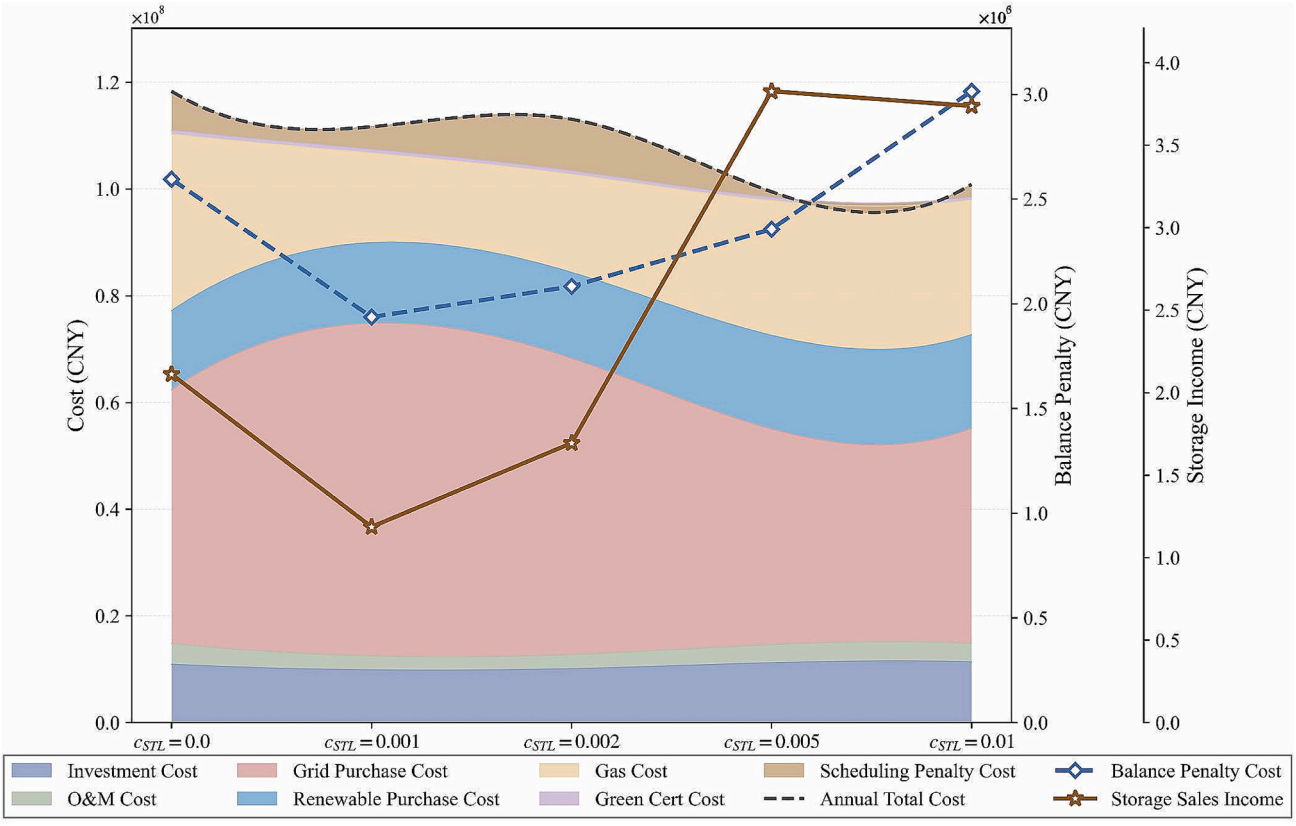

Fig. 16. Costs and revenues under different computing power scheduling costs.

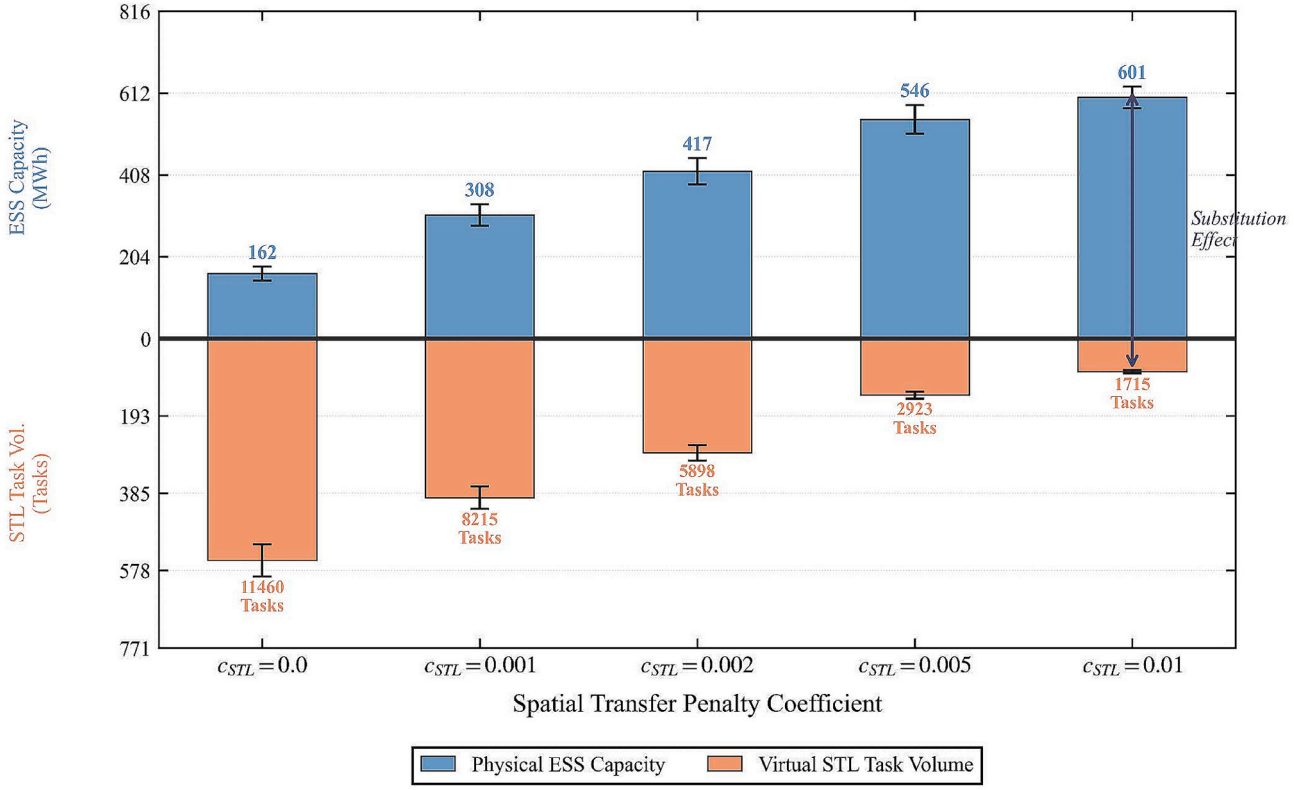

Fig. 17. Impact of computing power scheduling costs on flexible resource configuration results.

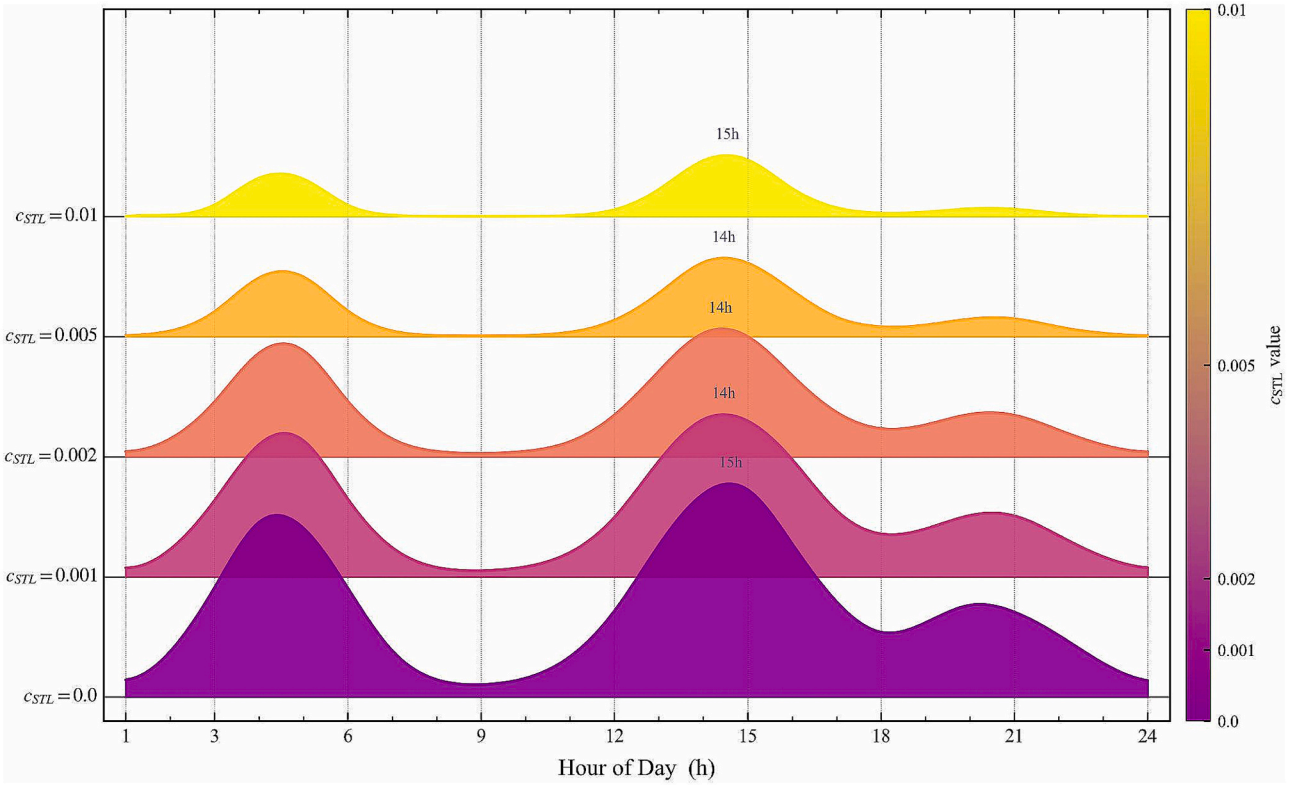

Fig. 18. Ridgeline plot of intraday spatial transfer volume under different scheduling costs.

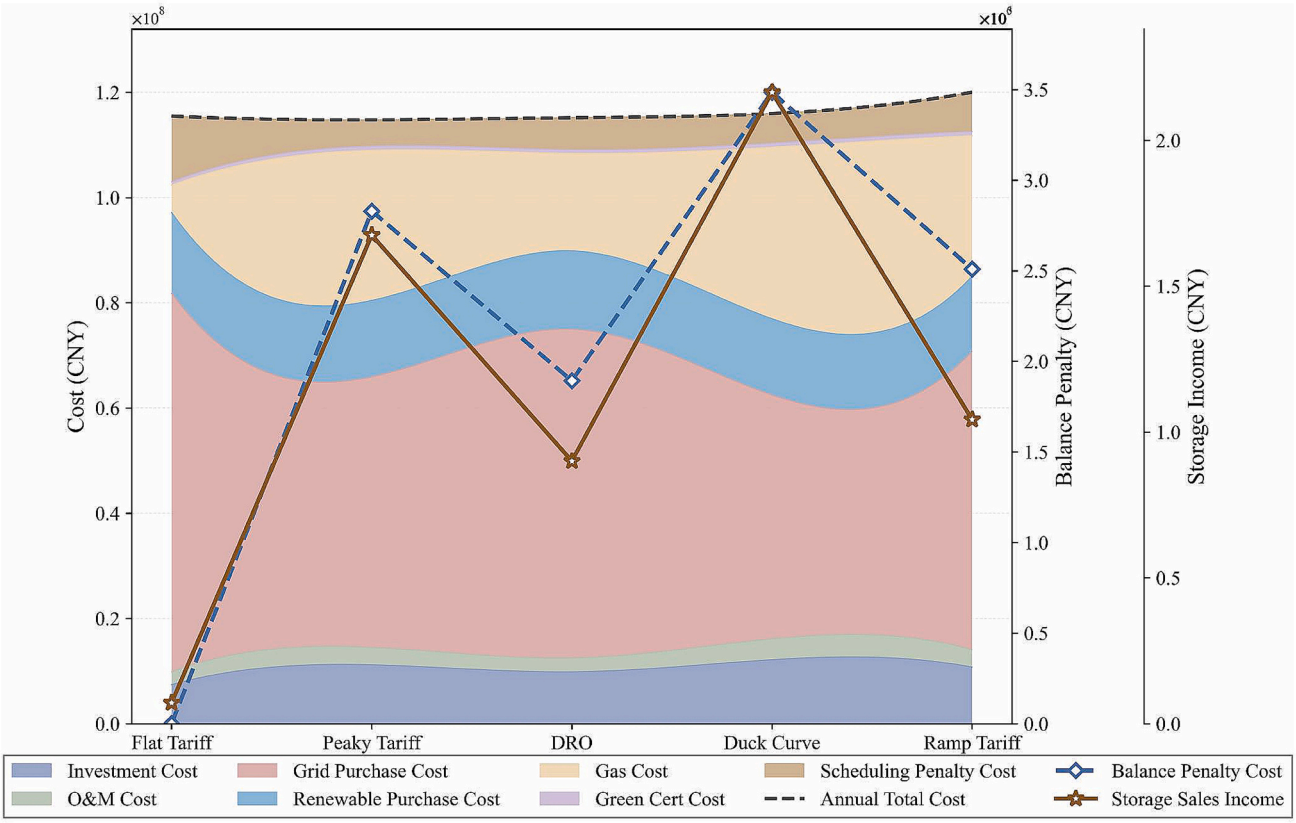

Fig. 19. Costs and revenues under different time-of-use electricity price curves.

penalty coefficient increases to 0.01, the economic advantage of workload transfer declines. To maintain supply-demand balance, the model replaces virtual flexibility with physical storage. Energy storage capacity increases to about 601 MWh, whereas STL decreases to approximately 1715 tasks. These results demonstrate that computing-electricity coupling can effectively reduce investment in physical flexibility resources.

The impact of transmission costs is further reflected in the intraday distribution of STL. The corresponding results are shown in Fig. 18.

The ridgeline plot shows that STL is mainly concentrated between 04:00–05:00 and 14:00–15:00. These periods coincide with high wind and photovoltaic generation, indicating that workload migration is primarily driven by renewable energy accommodation. As the space transfer penalty coefficient increases, STL decreases across all periods, and the distribution peaks become progressively flatter. Cross regional workload migration is therefore activated only when local energy imbalance becomes sufficiently severe to justify the transmission cost. This result demonstrates the sensitivity of the proposed planning model to economic signals.

5.4.4. *Analysis of time of use electricity pricing mechanism*

Time of use electricity pricing provides an important economic signal

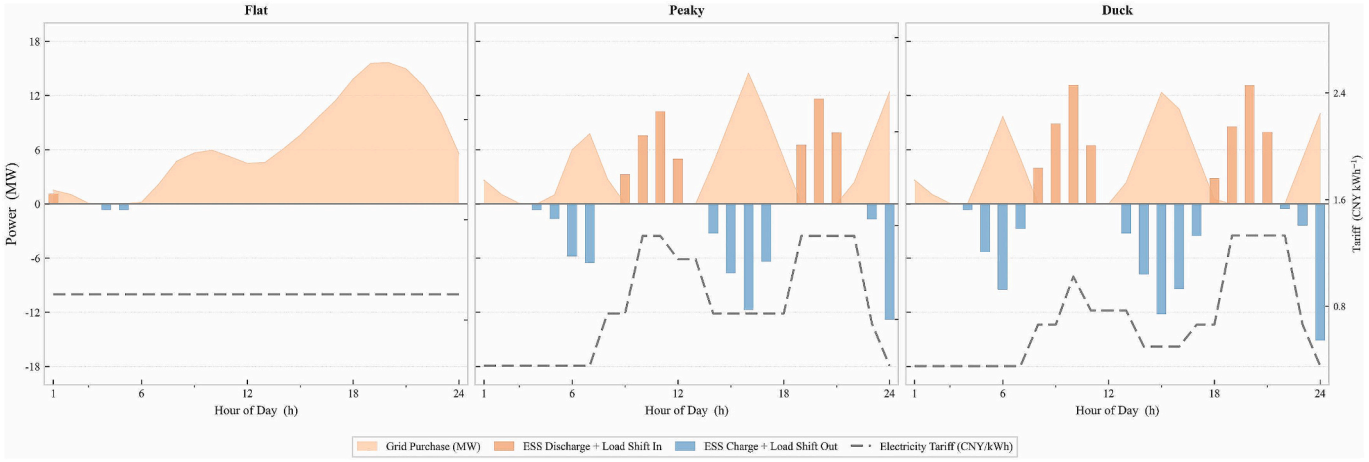

Fig. 20. Analysis of summer intraday operation under typical electricity price curves.

for activating workload flexibility and demand side response in DCC. To evaluate the influence of different pricing schemes, five scenarios are considered, including flat pricing, peak valley pricing, baseline DRO pricing, duck curve pricing, and linear ramping pricing. The corresponding electricity price profiles are presented in Appendix Fig. 3.

The electricity price profile directly affects the configuration of flexible resources. Under the duck curve pricing scenario, large photovoltaic generation at midday and rapid evening net load growth requires stronger ramping capability. Accordingly, the model installs the maximum number of CCHP units. Under the peak valley pricing scenario, the larger price difference encourages extensive energy storage deployment, with the storage capacity at several nodes reaching 20 MWh. The economic results under different pricing schemes are presented in Fig. 19.

The results further illustrate the economic response to different electricity price signals. The peak valley pricing scenario achieves the highest energy storage revenue, reaching approximately 25 million CNY through cross period charging and discharging, thereby reducing the total annual cost. Under flat pricing, the absence of price variation weakens storage arbitrage, resulting in the highest dependence on direct grid supply. Although the duck curve and linear ramping pricing scenarios better match renewable generation characteristics, their stronger ramping requirements increase balancing penalties and fixed operation and maintenance costs.

To further illustrate the operational response to different pricing mechanisms, typical summer operating conditions under flat pricing, peak valley pricing, and duck curve pricing are compared in Fig. 20.

The intraday operating profiles show clear responses of the computing-electricity coupling system to electricity prices. Under flat pricing, grid electricity purchase closely follows the load profile, while energy storage operation and workload migration remain limited. Under peak valley pricing, the system exhibits significant temporal workload flexibility. Energy storage charging and workload shifting are concentrated during low price periods, whereas energy storage discharging and workload execution are shifted toward peak price periods.

Under duck curve pricing, the midday price valley coincides with peak photovoltaic generation. The model therefore schedules intensive energy storage charging and workload execution during midday, followed by concentrated energy storage discharging during the evening peak price period. This strategy effectively mitigates grid ramping requirements. Overall, the results demonstrate that an appropriate time of use pricing mechanism can effectively activate workload flexibility in both temporal and spatial dimensions, thereby improving the coordination between DCC and the power grid while enhancing overall economic performance.

6. Conclusion

As the rapid growth of computing demand continues to increase electricity consumption in DCCs, computing-electricity coupling has become an important challenge for integrated energy system planning. This paper proposed a two-stage DRO bi-level planning model for DCC under supply-demand uncertainties. Through case studies and sensitivity analyses, the main conclusions are summarized as follows.

(1) Workload flexibility effectively mitigates the temporal mismatch between renewable generation and electricity demand. Under the computing-electricity coupling framework, TSLs are shifted to periods with abundant renewable generation, while STLs are migrated to regions with higher renewable availability. This coordinated scheduling strategy enables computing loads to provide equivalent flexibility for system regulation. Combined with the cascade energy utilization of the CCHP system, the proposed model reduces the total life cycle cost by 5.43% and decreases renewable energy curtailment by 1.18%. It also smooths net load fluctuations and reduces investment in redundant energy infrastructure.

(2) The proposed DRO framework achieves a better balance between planning economy and operational robustness. By constructing an ambiguity set based on joint moment information, the model captures supply-demand uncertainties without requiring explicit probability distributions. Compared with SP, the proposed method reduces the risk caused by distribution misspecification. Compared with conventional RO, it avoids excessive conservatism while maintaining the required reliability level. Numerical results show that the annual total cost is reduced by approximately 13% relative to the conventional two-stage RO model, while green certificate procurement costs are also effectively reduced.

(3) The coordinated configuration of multi-energy resources and the C&CG solution strategy ensures both planning effectiveness and computational efficiency. The complementary operation of CCHP units and energy storage improves overall energy utilization. Sensitivity analyses indicate that moderate renewable accommodation targets and appropriate CCHP capacity achieve a better tradeoff between economy and environmental performance, whereas overly stringent environmental constraints or excessive reserve capacity significantly increase life cycle cost. Meanwhile, the C&CG algorithm efficiently solves the two-stage Min-Max-Min problem through iterative interactions between the MP and SP, providing stable convergence within an acceptable computational time.

Future work will focus on further improving both the uncertainty modeling and market interaction mechanisms. Continuous time ambiguity sets will be investigated to better characterize long tail uncertainty beyond discrete scenario representations. In addition, electricity market dynamics, wide area communication delays, and carbon trading mechanisms will be incorporated to develop decentralized low-carbon planning and operation strategies for DCC under multi stakeholder coordination.

CRediT authorship contribution statement

Shuo Zhang: Writing – original draft, Methodology, Conceptualization. **Shaohui Ding:** Writing – original draft, Software, Investigation, Data curation. **Yingzi Li:** Writing – review & editing, Methodology. **Ming Wei:** Visualization, Formal analysis, Data curation. **Qinliang Tan:** Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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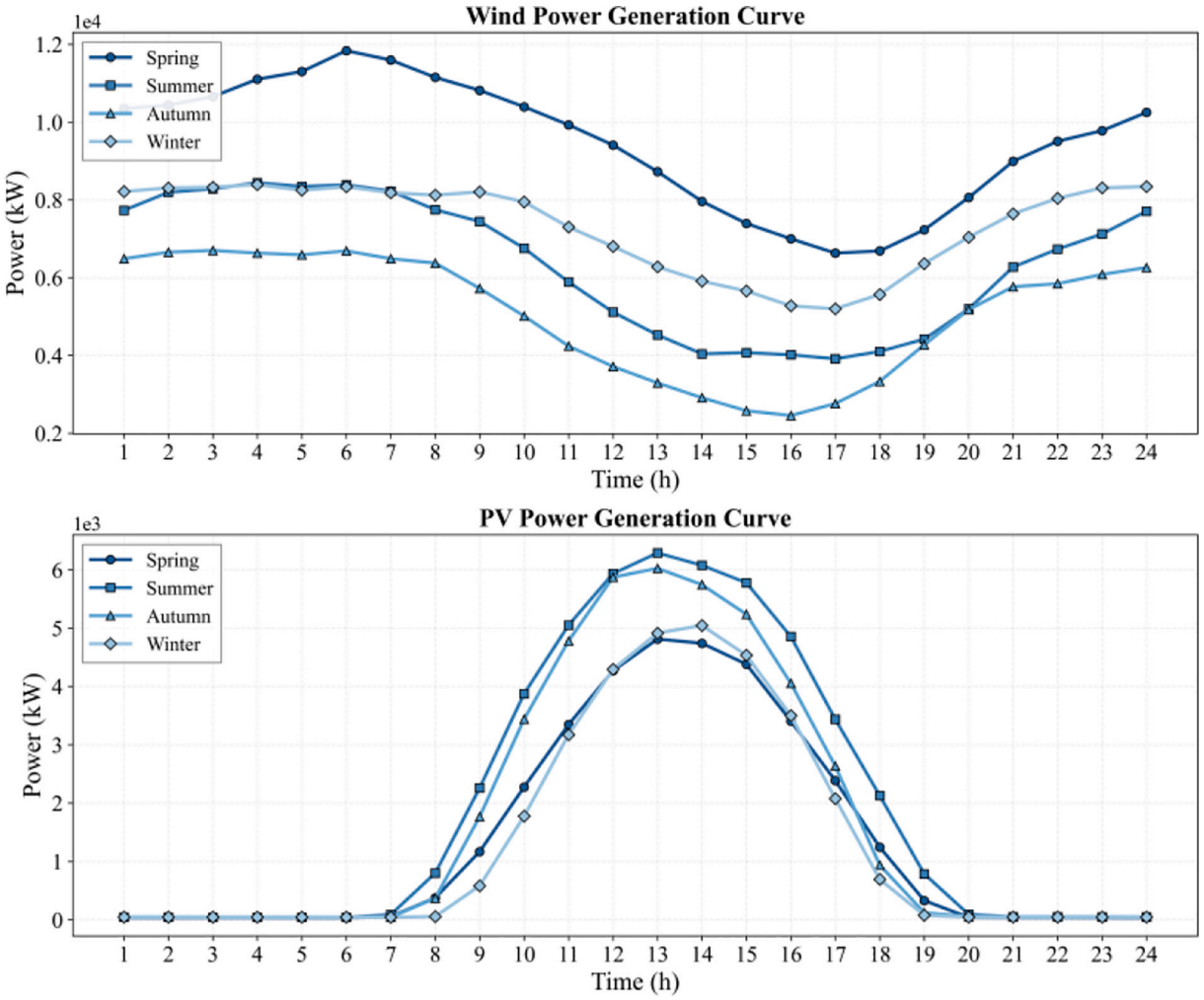
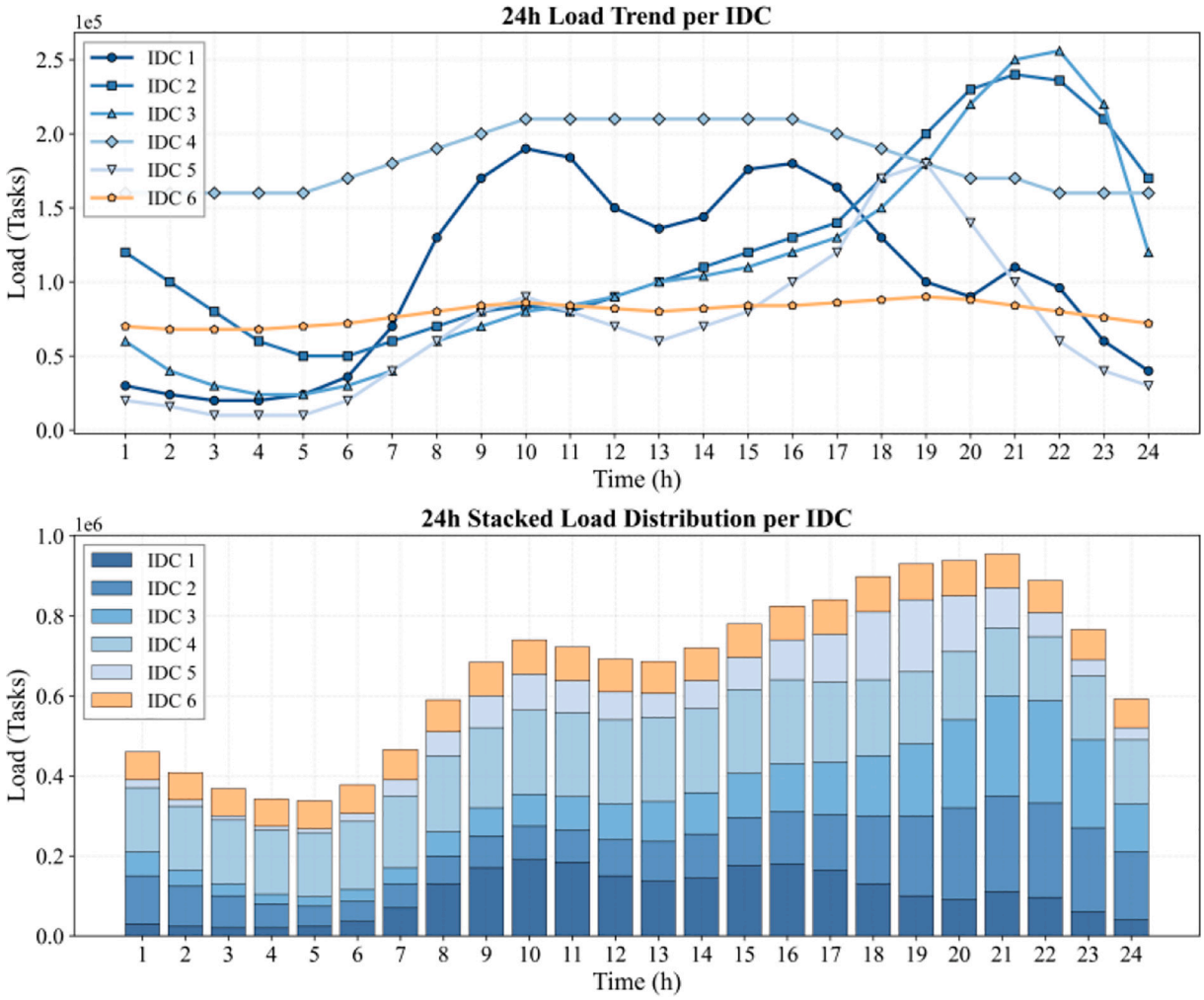
Appendix A. Appendix

The actual wind and photovoltaic power output curves for typical days in each season used in the case study are shown in Appendix Fig. 1. These data illustrate the temporal variation of renewable generation under different seasonal and meteorological conditions.



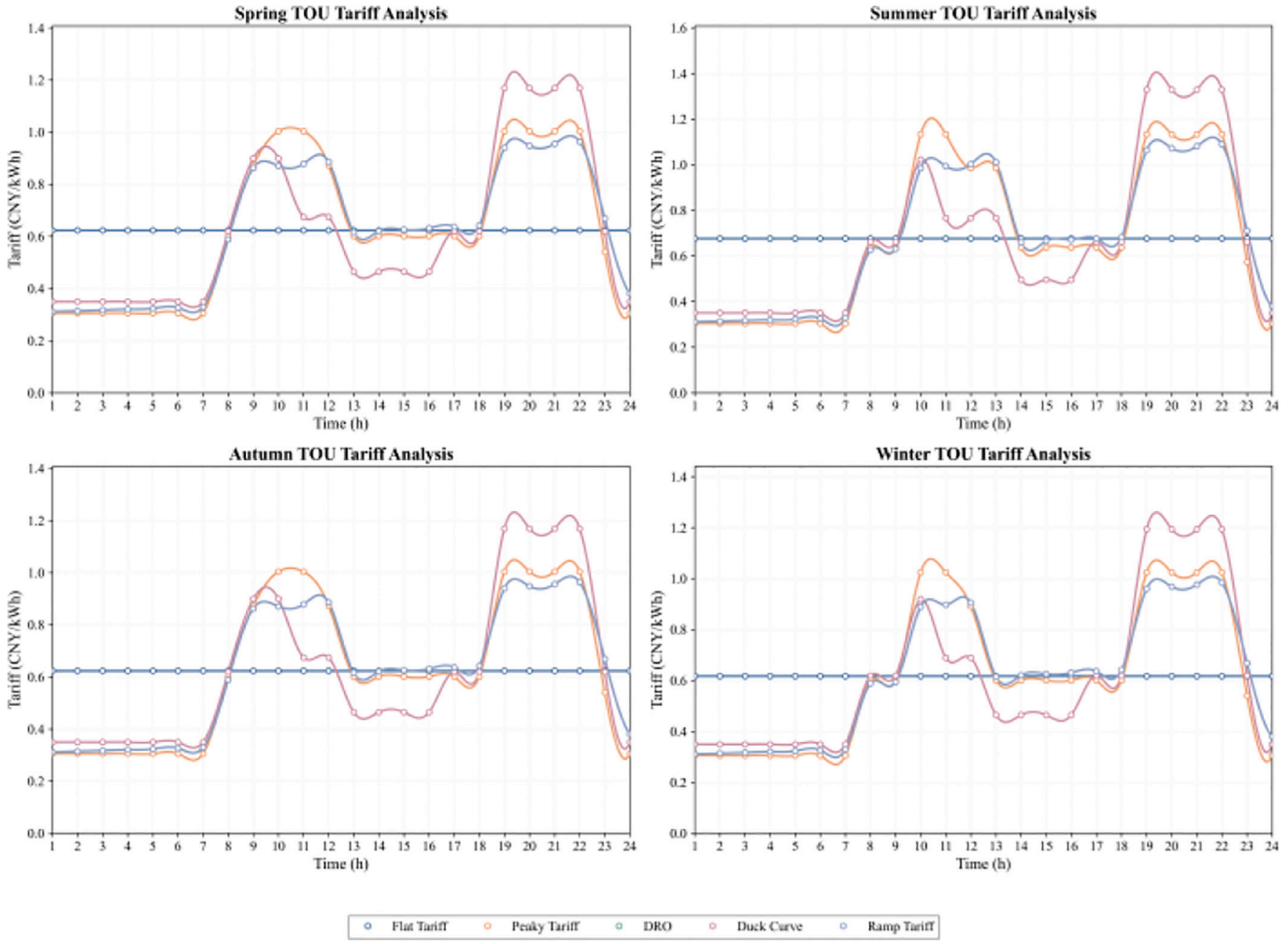
Appendix Fig. 1. Renewable energy generation power curves.

To coordinate computing resources with renewable energy supply, the DCC includes six types of DCs, namely e commerce, gaming, finance, cloud services, streaming media, and the Internet of Things. The arrival profiles of computing tasks for each DC on typical days are shown in Appendix Fig. 2.



Appendix Fig. 2. Load task volume of each data center.

The time series electricity price profiles used in the sensitivity analysis, including flat pricing, peak valley pricing, duck curve pricing, and linear ramping pricing, are shown in Appendix Fig. 3.



Appendix Fig. 3. Time-of-use electricity price scenario curves.

The economic evaluation of system capacity planning depends on the investment and operating parameters of the physical equipment. The unit construction costs of different DCs in the DCC are listed in Appendix Table 1. The operating limits and techno economic parameters of the remaining equipment, including CCHP units, energy storage systems, servers, and cooling systems, are summarized in Appendix Table 2.

Appendix Table 1
Unit construction cost of different data centers (CNY/㎡).

IDC1	IDC2	IDC3	IDC4	IDC5	IDC6
12,000	10,000	16,000	8000	15,000	7000

Appendix Table 2
Technical and economic parameters.

Storage system parameters		R	0.0018 (°C/kW)
ρ	0.15	μ_{ER}^{COP}	5
η_f	0.95		
η_c	0.95	**CCHP system parameters**	
c_{cn}^{yw}	10 (kWh)	H_G	10 (kWh/m^3)
		P_{PGU}^{max}	2000 (kW)
Load scheduling parameters		c_p	4.2
M	8 (G)	ε_{PGU}	0.35
c_{VM}	1200 (CNY/set)	m_{HP}	20
		μ_{HP}^{COP}	4
IDC parameters		c_g	0.44 (CNY/kWh)
$P_{IT,free}$	0.4 (kW)	η_{PGU}	0.4
θ	20 (G/h)	μ_{AR}^{COP}	1.2
C	1/R	m_{AR}	10

(*continued on next page*)

Appendix Table 2 (*continued*)

ε	0.9	T_{AR}^{max}	60 (°C)
c_{IDC}^{yw}	0.05 (CNY/kW)	ε_{HP}	0.2
$P_{IT,full}$	0.95 (kW)	c_{CCHP}	4,600,000 (CNY)
δ_{PDU}	0.8	c_{PGU}^{yw}	0.05 (CNY/kW)

Data availability

Data will be made available on request.

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